

Vegetation-induced modifications in hydrological processes and the consequential dynamic effects of slope stability

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ARTICLE INFO

Keywords:

Hydrological processes
Slope stability
Vegetation
Landslides
Heterogeneity

ABSTRACT

Vegetation, as a crucial factor influencing the occurrence of shallow landslides, faces increased demands for understanding how its hydrological effects impact landslide processes and mechanisms due to the rise in extreme rainfall events. Vegetation-induced modifications (VIMs), such as transpiration, rainfall interception, and root channels, affect various hydrological processes (e.g., infiltration, preferential flow, or stemflow) through different mechanisms and exhibit high temporal dependency. The aforementioned vegetation hydrologic effects dynamically strengthen or weaken the soil, increasing the uncertainty of the overall slope stability in time and space. Therefore, a comprehensive and systematic understanding of these dynamic processes is essential for accurately predicting the timing and location of shallow landslides. This review first examines how various forms of VIMs influence slope surface and soil hydrological processes, summarizing the associated parameters to establish a theoretical framework for the intrinsic link between VIMs and slope hydrological processes. Subsequently, we examine the impact of each vegetation-induced hydrological process on the temporal and spatial variations of local soil strength and overall slope stability under rainfall conditions. Using a space-time cube model, we develop a framework to elucidate when and where vegetation-induced hydrological processes exert positive or negative effects on slope stability. This review contributes to a deeper understanding of the complex interactions between vegetation, soil hydrology and slope stability, and proposes a new theoretical framework for dynamic modeling of slope stability based on complex hydrological processes.

1. Introduction

With the rapid changes in the terrestrial water cycle and the increasing frequency of extreme rainfall events, the role of hydrology in natural hazards is being redefined (Coumou and Rahmstorf, 2012; Hallegatte et al., 2013; Lehner et al., 2017; Vereecken et al., 2022). Rainfall-induced shallow landslides are among the most catastrophic hazards due to their sudden onset, long travel distances, and high velocities (Iverson, 2000). Furthermore, the frequency of extreme rainfall events has been closely monitored and shown to correlate with the probability and magnitude of rainfall-induced landslides (Chen and Sun, 2009; Crozier, 2010; Gariano and Guzzetti, 2016; Gariano et al., 2017; Maragaño-Carmona et al., 2023; Sidle and Bogaard, 2016).

Understanding how rainfall-induced hydrological changes in the slope surface and soil affect slope stability has become critical for predicting and preventing such potential landslides.

As the natural cover material on slopes, vegetation has a significant influence on water movement, changes in soil strength, and the occurrence of shallow landslides during rainfall events. The way plants reshape surface and near-root soil hydrological processes is a multi-mechanism issues, and these processes are key to understanding how plants influence soil structure in fine scale and infiltration processes and stability in slope scale (Vereecken et al., 2022). The effects that influence both surface and subsurface hydrological processes, such as transpiration, rainfall interception, and root channels, are referred to as vegetation-induced modifications (VIMs). Vegetation transpiration,

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<https://doi.org/10.1016/j.catena.2025.108793>

Received 30 August 2024; Received in revised form 25 January 2025; Accepted 28 January 2025

Available online 17 February 2025

0341-8162/© 2025 Published by Elsevier B.V.

respiration, and photosynthesis can reduce soil saturation in shallow slope soils under wet conditions or after rainfall (Laio, 2006; Ng et al., 2022). Under dry conditions, however, these processes can cause soil cracking and increase soil permeability (Anderson et al., 1982; Coppin and Richards, 1990). The root system increases soil water retention within a certain range, but also increases the heterogeneity of soil structure and the differences in hydraulic conductivity within the slope (Scanlan, 2009). For woody vegetation, canopy interception can prevent or delay infiltration of some rainfall (Gonzalez-Ollauri and Mickovski, 2017a), but it can also lead to stemflow and water accumulation around roots (Hanchi and Rapp, 1997; Liang et al., 2011). For herbaceous vegetation, canopy can intercept the amount of rainfall, but its surface-mat effect can reduce surface runoff velocity (Gray, 1995; Morgan, 2007), thereby affecting the amount of rainfall infiltration. The aforementioned processes result in a more dynamic assessment of slope stability. The generation of seepage force and the additional weight of rainfall contribute to sliding forces, while vegetation affects sliding and anti-sliding forces through its influence on infiltration rate and instantaneous seepage force. The hydrological effects of vegetation, by altering infiltration rates and reshaping soil structure, influence the distribution of pore water pressure and cohesion within the slope. Additionally, soil structure influenced by root activity creates preferential hydraulic pathways, resulting in the accumulation or dissipation of pore water pressure (Löbmann et al., 2020; Ng et al., 2022).

This complex variation in the representation of VIMs in fine scale or monoculture hydrological processes affects soil strength and overall slope stability. At the basin or catchment scale, such complex processes result in slope units with similar landslide-predisposing factors (i.e. soil type, slope, elevation, vegetation cover, etc.) showing differentiation between destabilization and stabilization. Study in the Loess Plateau of China showed that vegetation facilitates shallow landslides during prolonged rainfall periods due to differences in soil properties between the surface root-soil complexes and the underlying loess in this region, as well as the presence of preferential flow along root channels (Guo et al., 2020b). Lin et al. (2024) have shown that typical clusters of vegetation-covered landslides in the Wuping area of Fujian Province, China, demonstrate that lateral slopes with similar vegetation cover would be subject to different levels of water loss due to differences in the distribution of the root system. The study in Mengdong, Yunnan Province, China, showed that the concentration of landslides was higher on slopes with higher NDVI (Hu et al., 2024). The complex divergence phenomenon of typical clusters of vegetated shallow landslides has received more attention in recent years.

Previous studies have either considered simplified hydrological effects in slope stability analysis, neglecting the various mechanisms by which vegetation alters hydrological processes (Bittelli et al., 2012; Godt et al., 2009; Li et al., 2021; Ng et al., 2021), or have focused on specific hydrological processes, thereby simplifying slope-scale mechanisms by primarily attributing landslide occurrence to reduced matric suction (Boldrin et al., 2017; Hawke and McConchie, 2011; Liu et al., 2022; Osman and Barakbah, 2006; Reubens et al., 2007; Wright et al., 2007). Although these studies have increased the interest in VIMs in hydrological processes and slope stability, there is still a lack of in-depth studies on the positive or negative impacts of VIMs in hydrological processes and slope stability, which would be the basis for revealing the spatial and temporal distribution of clustered rainfall landslides.

Therefore, to elucidate the mechanisms of VIMs in hydrological processes and to accurately predict the spatiotemporal occurrence of shallow landslides, a comprehensive and systematic review addressing the multiple processes and mechanisms of VIMs in hydrological processes and their effects on slope stability is needed. This review aims to integrate current research on VIMs in hydrological processes by various means and to assess the magnitude and positive or negative effects of these hydrological effects on slope stability under complex dynamic conditions. A systematic framework for delineating the intrinsic relationships between VIMs and hydrological processes is proposed. Based

on this framework and the concept of space-time cubes, a conceptual model of the dynamic effects of slope stability to VIMs in hydrological processes at the slope scale is proposed to provide new insights for future research on rainfall-induced landslides.

2. Vegetation-induced modifications in hydrological processes

In contrast to the few slopes that are defined by unstructured soil and uniform topography, the majority of natural slopes exhibit intricate internal structures and varied terrain. The permeability of the soil matrix within the slope's interior can be employed to quantify the partial subsurface water flux. However, it is also essential to consider the permeability within fractures and microporous spaces, as they play a crucial role in slope hydrological processes. The occurrence of preferential flow in these fractures and macropores significantly impacts pore water pressure and can affect the timing of slope failure (Malet et al., 2005; Van Beek and Van Asch, 2005; van der Spek et al., 2013; Wienhöfer et al., 2011). Furthermore, surface geomorphic processes exert a significant influence on the runoff and infiltration of rainfall across the slope surface. The involvement of vegetation introduces a further layer of complexity to the study of hydrological processes. Within the vegetation, bottom-up hydrological pathways are formed. Underground, the presence of vegetation serves to accentuate the delineation of areas of preferential flow and areas of substrate infiltration. Aboveground, vegetation affects runoff and creates stem flow and throughfall, hence, infiltration conditions become more heterogeneous (see Fig. 1).

2.1. Patterns of vegetation-induced modifications in hydrological processes

The slope hydrological processes can be divided into three categories: intraplant hydrological processes, underground hydrological processes, and aboveground hydrological processes.

2.1.1. Intraplant hydrological process

Plants employ the xylem and phloem to facilitate the transfer of water from the soil to the atmosphere via their leaves (Bordoloi and Ng, 2020), opening up a new pathway for water movement from the soil to the atmosphere (Liu et al., 2022). Root hairs, particularly in herbaceous vegetation, exhibit the highest permeability and serve as the primary pathway for water entry into plants from the soil (Ng et al., 2022), as illustrated in Fig. 2(e). Subsequently, water moves through the epidermis, cortex, and endodermis before gradually entering the xylem vessels. The pathways for water movement include (Maurel et al., 2008; Steudle, 2000): 1) apoplastic pathway surrounding the protoplasts, 2) symplastic pathway through plasmodesmata and shared cytoplasmic components between cells, and 3) transcellular pathway across cell membranes. These pathways are illustrated in 2(d). Due to the presence of a reverse hydraulic gradient, water gradually moves upward in the xylem after entering it, as shown in Fig. 2(c). A portion of the water is consumed through photosynthesis and respiration, while the remaining portion is released into the atmosphere through transpiration (Aroca et al., 2012; Daly et al., 2004a, 2004b), as shown in Fig. 2(b). It should be noted that the xylem functions as a porous medium (Wheeler and Stroock, 2008), and therefore, water transport within plants, from the root system to the canopy, follows Darcy's law.

The intraplant hydraulic pathways expand the drainage interface of the soil and exhibit higher drainage rates compared to bare soil. During drought periods, the transpiration loss of moisture in vegetated areas is greater than the evaporative loss on the surface of bare soil (Vereecken et al., 2022). In particular, deep-root systems play a role in hydraulic lift, absorbing moisture from deeper soil layers, enabling adjacent deep soil layers near the roots to participate in drainage at an earlier stage.

• Consumption and release

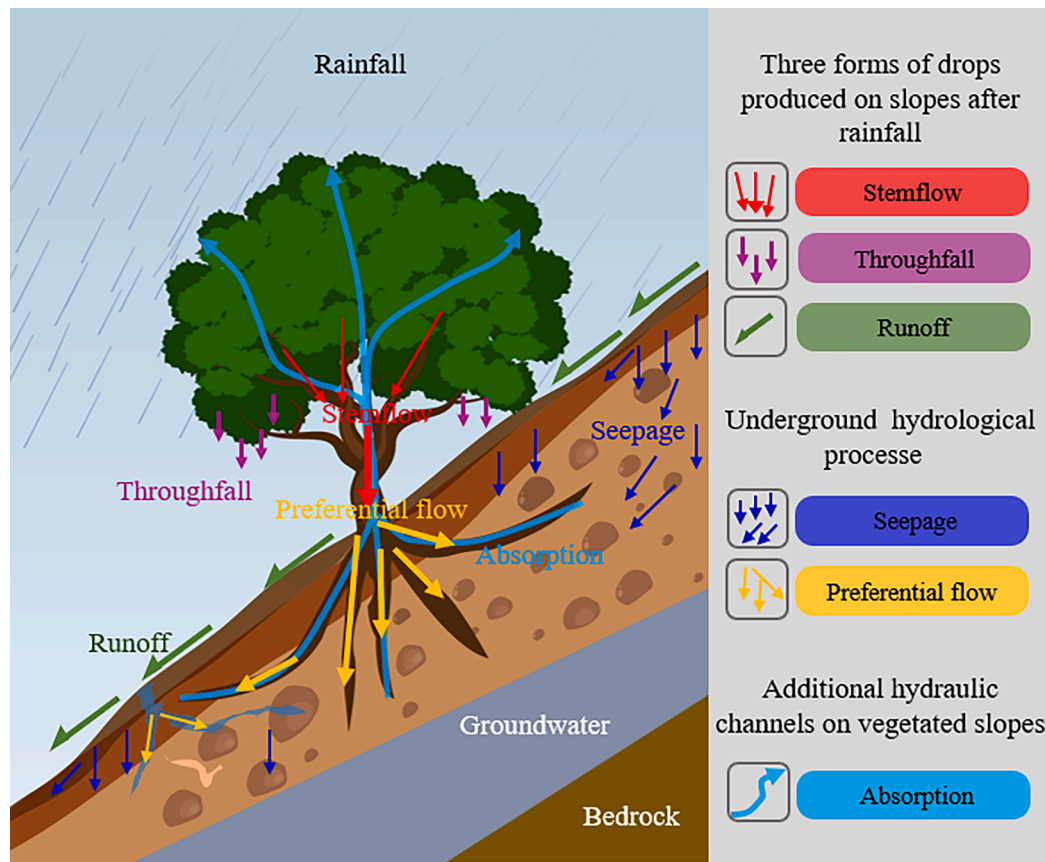


Fig. 1. A schematic representation of the role of VIMs in hydrological processes.

The VIMs that significantly impact the internal hydrological processes of plants can be mainly categorized into water release caused by transpiration and water consumption due to photosynthesis and respiration (Brodribb and Cochard, 2009; Cai et al., 2022; Vereecken et al., 2022; Wong et al., 1979). Research suggests that plants can use up to 95 % of the water absorbed by their root systems through transpiration (McElrone et al., 2013).

Transpiration rates are affected by solar radiation, humidity, and the opening and closing of leaf stomata (Ng et al., 2022). During rainfall events, transpiration efficiency (TE) decreases (Aparecido et al., 2016; Hadiwijaya et al., 2020; Yasutake et al., 2018). Solar radiation is lower at night, and transpiration rates when stomata are open for respiration are typically 5 %–15 % of daytime transpiration rates (Caird et al., 2007; Dawson et al., 2007). Although the amount of water consumed by vegetation itself is relatively small compared to root water uptake, it is closely related to the rate of transpiration. During photosynthesis, plants open stomata to take up carbon dioxide, thereby increasing transpiration rates and accelerating root water uptake, as shown in Fig. 2(b). During the drying phase after rainfall, vegetation first absorbs water from the rhizosphere and the preferential flow network connected to it, creating a negative pressure gradient around it, causing the surrounding matric water to gradually move towards the preferential flow network, and the water taken up by the roots is eventually consumed or released.

As a result, the soil near the roots can maintain relatively low saturation and high matric suction (Lim et al., 1996; Simon and Collison, 2002). On the other hand, vegetation promotes soil moisture cycling, and excessive evaporation during dry periods can cause root decay, soil cracking and the formation of new shrinkage cracks, while high soil moisture during heavy rainfall can cause soil softening and closure of pre-existing shrinkage cracks (Anderson et al., 1982; Coppin and Richards, 1990; Johnson, 2018). Thus, under extreme conditions, these VIMs alter the hydraulic efficiency within the vegetation, thereby

affecting underground hydrological processes.

2.1.2. Underground hydrological processes

During rainfall events, the remaining rainwater falls directly onto the slope surface in addition to being intercepted or retained (Liang et al., 2011). Some of the rainwater infiltrates directly into the soil, while some forms runoff on the slope accompanied by infiltration. As the rainwater continuously infiltrates vertically, it initially fills the top soil layer, causing the soil moisture content to increase and even reach the saturation stage. As the wetting front advances, there is a gradual transition from unsaturated to saturated conditions and even complete saturation from shallow to deep areas (Lumb, 1975). The saturated zone expands and loses suction, while the unsaturated zone reduces and experiences a decrease in suction (Capparelli and Versace, 2014; Iverson, 2000; Torres et al., 1998). Because the permeability coefficient in the saturated zone is constant, while the permeability coefficient in the unsaturated soil is a function of soil suction (Gonzalez-Ollauri and Mickovski, 2017a; Ng and Pang, 2000; Scanlan, 2009), overall infiltration rates on slopes vary with the duration of rainfall events (Sidle and Bogaard, 2016).

However, most potential landslide-prone areas exhibit a highly complex network of rainfall infiltration and flow at the slope scale (Sidle and Bogaard, 2016). Therefore, in addition to Darcy's law for soil matrix flow (seepage), underground hydrological processes involve preferential flow that may defy classical models (Iverson, 2000). Preferential flow pathways represent regions in the soil where relatively rapid water and solute migration can occur during heavy rainfall or flooding (Perillo et al., 1999). A variety of structures coexist in underground, including root channels, animal burrows, soil cracks, macropores, impermeable layers, interfaces between soil layers, soil–bedrock interfaces, and fractures within weathered bedrock (Fan et al., 2020). Together, these structures and interfaces form a complex interconnected network that provides potential pathways for preferential flow. Rainwater

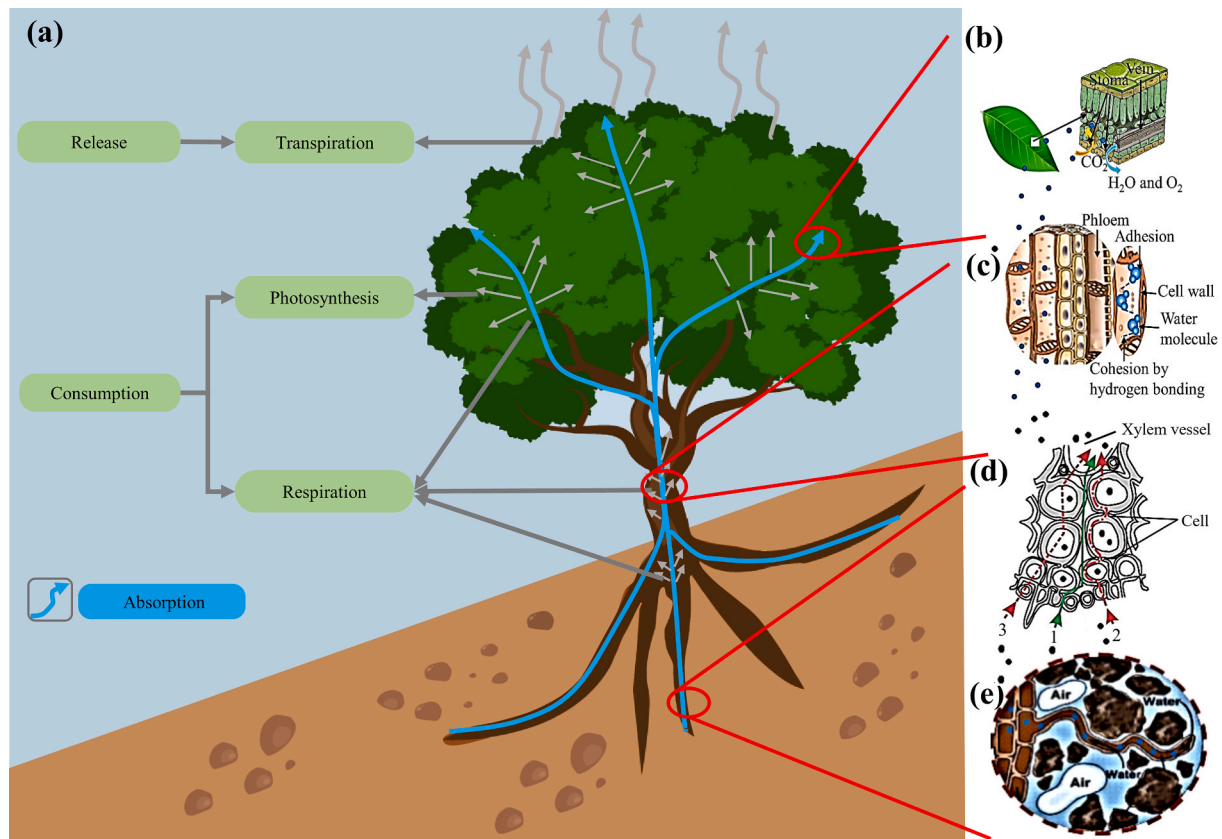


Fig. 2. Intraplant hydrological processes and release and consumption caused by vegetation functions. (a) Formation of hydraulic pathways within vegetation from roots through stems to leaves. Respiration of roots and stems consumes some water, while the remaining water is either consumed or released by leaves. (b) End point of hydraulic pathways within vegetation (Xiao et al., 2024). (c) Xylem transport in stems (Xiao et al., 2024). (d) Schematic representation of water and solute transport pathways from soil to root xylem (Steudle, 2000). (e) Three ways in which water enters the xylem (Ng et al., 2022).

preferentially infiltrates through soil fractures and macropores, preventing rapid wetting of smaller pores in the surrounding matrix (Beven and Germann, 2013). The larger pore size and greater tortuosity of macropores exhibit significant distinctions from the smaller pores within the surrounding matrix, resulting in higher hydraulic conductivity and pronounced non-uniform flow (Jarvis, 2007). Preferential flow can have a greater influence on local water transport and pore water pressure compared to seepage within the soil.

What is the role of vegetation in soil matrix and preferential flow processes? In addition to the consumption and release described above, root-induced modifications (RIMs) also significant impact on soil hydrological processes (Leung et al., 2015). Although limited research suggests that root penetration creates new preferential flow paths (Fageria and Stone, 2006; Ghestem et al., 2011), this process alters the hydraulic properties of the soil matrix due to squeeze connectivity and root exudates (Bacq-Labreuil et al., 2018; Bodner et al., 2014; Grayston et al., 1997; Hayashi et al., 2006; Scanlan, 2009; Scholl et al., 2014; Traoré et al., 2000).

• Squeeze and connectivity

Vegetation has the ability to induce soil compaction or loosening through its root system (Lu et al., 2020; Qin et al., 2022), thereby amplifying spatial heterogeneity within the root-influenced area (Clark et al., 2003; Scholl et al., 2014). The alterations in pores and fractures have varying degrees of impact on both matrix and preferential flow. In soil matrix, roots grow into pre-existing pores (Gish and Jury, 1983), can block hydraulic pathways (Meek et al., 1990; Murphy et al., 1993), and cause larger pores to split into smaller ones (Scanlan, 2009), leading to compaction in the vicinity of roots, as shown in Fig. 3d and e. This

compaction results in increased soil density and decreased permeability. However, coarse root growth can also generate new pores and microcracks in the surrounding soil, increasing pore size heterogeneity with more large and small pores and fewer medium pores (Scholl et al., 2014). Furthermore, the “bottleneck effect” suggests that larger soil pores and microcracks have higher permeability (Beven and Germann, 2013), thereby significantly improving permeability even when soil porosity remains largely unchanged. In addition, biological macropores and root-induced micropores are formed after root decay or degradation (Cresswell and Kirkegaard, 1995; Ghestem et al., 2011; Horn and Smucker, 2005; Mitchell et al., 1995; Wuest, 2001). Pores that were previously sealed by the presence of roots can become connected as roots die, increasing permeability (Gish and Jury, 1983; Murphy et al., 1993; Pagliai and Denobili, 1993; Whalley et al., 2005).

• Root exudates

Root exudates have a lubricating effect during root penetration of the soil matrix (Bais et al., 2006), facilitating the occurrence of the above processes. Aggregate stability refers to the percentage of stable aggregates in the soil after wetting (LeBissonnais, 1996), which serves as a crucial indicator of the soil’s resistance to wetting-induced dispersion and controls soil permeability (Bryan, 1968). Root exudates influence microbial activity (Czarnes et al., 2000) and result in: the production of glomalin-related soil protein (GRSP), which is positively correlated with aggregate size (Wright et al., 2007), and varying degrees of enhancement of interparticle adhesion and soil hydrophobicity by different exudates (Barré and Hallett, 2009; Czarnes et al., 2000; Peng et al., 2011). In summary, root exudates improve aggregate stability (Fattet et al., 2011; Frei et al., 2003; Gyssels et al., 2005; Jastrow et al., 1998), reduce

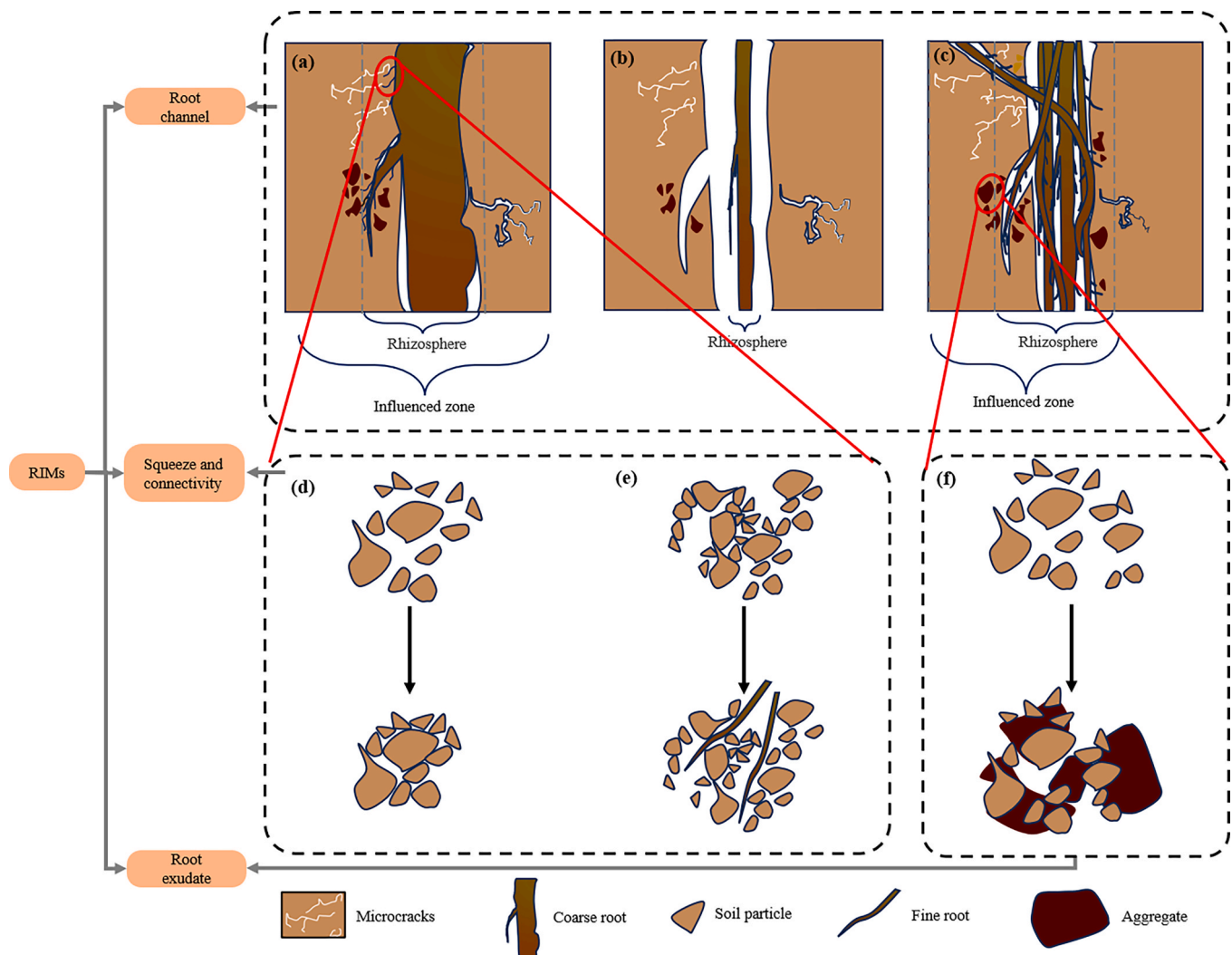


Fig. 3. Underground hydrological processes caused by RIMs. (a) Channels occupied by native roots, (b) almost empty channels occupied by decomposed roots, and (c) channels originally empty or occupied by decomposed roots and later occupied by new roots (Ghestem et al., 2011). (d) Compression of the soil around the root channel. (e) Fine roots connect the cracks that were already exist. (f) Effect of root exudates.

wetting-induced dispersion during rainfall infiltration, and to some extent, inhibit changes in soil structure and porosity, as shown in Fig. 3 (f).

• Root channels

Vegetation has the ability to modify existing potential preferential flow paths or create new ones. Studies have shown a strong correlation between macropores and roots (Stewart et al., 1999). Macropores formed by root channels account for more than 55 % of all macropores (Noguchi et al., 1997; Perillo et al., 1999), making them the primary form of vegetation structure influencing preferential flow paths. The presence of root channels is closely related to roots, and the structure and function of the rhizosphere influences the formation of root channels. The rhizosphere refers to the collective composition of roots, root sheaths (if present), microorganisms, and the surrounding soil (Tarafdar and Jungk, 1987), which has different physical, chemical, and biological properties compared to non-rhizosphere soil. The rhizosphere contains higher levels of organic matter due to root and microbial secretions, which facilitates stability and aggregate formation (Fageria and Stone, 2006), making the root channels and surrounding soil structure more stable. Root systems undergo a series of processes, such as growth and decay, expansion and contraction, and aggregation, which result in the

formation of root pores, macropores, and microcracks (Beven, 1981; Beven and Germann, 1982; Coppola et al., 2009). Lignin, the structural framework of wood responsible for water and nutrient absorption (Khalil et al., 2015; Reddy and Yang, 2005), causes woody roots to increase their volume and increase the diameter of root channels through the accumulation of woody tissue (Ghestem et al., 2011). In addition, root compression and root hair invasion cause rearrangement of the surrounding soil, altering the hydraulic conductivity of the soil matrix and potentially creating new cracks and macropores (Fig. 3(a)). In addition to volume changes due to growth, changes in soil moisture content also affect root volume. The volume of lignocellulose decreases during dry periods and increases during wet periods (Bordoloi et al., 2017). Water uptake and root expansion further compresses the surrounding soil and enlarges root pores. However, during shrinkage due to water deficit, roots may separate from the pore walls, creating temporary preferential flow paths until the roots expand again upon the next rainfall event (Fig. 3(b)). Root degradation and decay can also result in gaps between roots and pore walls, exacerbating the development of localized pore pressure (Mitchell et al., 1995; Perillo et al., 1999). In addition, roots tend to aggregate within macropores (Pierret et al., 1999). White and Kirkegaard (2010) found that 90 % of wheat roots grow within existing pores in deeper soil layers. The easy accessibility and low resistance to water and air in macropores may be the reason

why root systems (native or invasive) aggregate in these areas. Fine roots exert less stress on soil particles, making them difficult to displace (Bengough et al., 2011). Therefore, after experiencing external forces, fine roots tend to bend and aggregate within existing pores (Fig. 3(c)). If the pore walls are weak, roots present in the soil matrix or other larger pores may invade the channels. The presence of large numbers of roots reduces the hydraulic conductivity of macropores, which may be even lower than that of the soil matrix (Kar and Ghildyal, 1975; Meek et al., 1990; Murphy et al., 1993). For herbaceous plants, the surface-mat effect (Coppin and Richards, 1990) they create may be a specific manifestation of how root aggregation affects soil hydraulic conductivity. The densely woven plant fibers grow or invade the macropores in shallow soil, “stitching” potential preferential flow paths (Löbmann et al., 2020).

Based on these effects, we find that vegetation exhibits a high degree of variability in soil hydraulic conductivity. Some scholars argue that roots promote soil loosening, thereby increasing the porosity and permeability of shallow soil layers (Dunkerley, 2002; Seyfried, 1991; Vergani and Graf, 2016). Conversely, others suggest that vegetation can prevent surface soil erosion and deformation, thereby maintaining soil porosity and permeability (Coppin and Richards, 1990; Gray, 1995). Additionally, some researchers suggest that root proliferation in stressed soils can cause compaction around roots, ultimately reducing soil porosity (Ni et al., 2019; Singh et al., 2018). Scanlan (2009) found no significant effect of vegetation on soil hydraulic properties in their experiments with sandy soils. Malet et al. (2005) reported that herbaceous vegetation had a minimal effect on pore water pressure. On the other hand, Löbmann et al. (2020) found both positive and negative effects of herbaceous vegetation on soil hydrological conditions. These results may be attributed to spatiotemporal effects and the combined positive and negative influences of multiple factors. Therefore, a quantitative explanation of how vegetation modifies soil permeability requires a comprehensive consideration of multiple influencing factors and spatiotemporal effects.

Moreover, the current description and prediction of infiltration dynamics in soils with hidden underground structural features are insufficient to fully comprehend the actual infiltration processes (Fan et al., 2020; Lin, 2010). Classical infiltration models, such as Darcy’s law and the Richards equation (Richards, 1931), typically assume homogeneous and uniform flow in the medium. However, this assumption rarely holds in highly heterogeneous underground structures (Lin, 2010). Roots can either impede or facilitate the development of preferential flow channels, and many studies have shown that root infiltration contributes more to pore formation than to pore plugging (Bogner et al., 2010; Sullivan et al., 2022). This, of course, depends on the proportions of the three root channels mentioned above and initial soil moisture conditions. Some root-generated potential preferential flow paths exhibit high hydraulic conductivity. However, due to the presence of other preferential flow paths, water may bypass these macropores and be redirected to areas with lower hydraulic gradients. The influence of root channels on soil hydrological processes becomes significant only under extreme rainfall conditions. Therefore, the main challenge in incorporating root channels and their secondary pathways into hydrological models is the highly variable parameterization of root effects.

2.1.3. Aboveground hydrological processes

Vegetation not only alters hydrological processes within the soil, but also influences the temporal and spatial distribution of precipitation, thereby affecting aboveground hydrological processes. After rainfall occurs, the canopy and branches of woody plants absorb and evaporate a portion of the rainfall, known as interception loss, while the remaining water eventually reaches the ground as throughfall or stemflow, as shown in Fig. 4. Throughfall refers to the portion of rainfall that drips down the slope in a sheet-like form, while the remaining rainwater is collected at the top of tree trunks and delivered to the slope in a concentrated form known as stemflow (Pinos et al., 2022). Throughfall and stemflow buffer rainfall and tend to collect around base of trees,

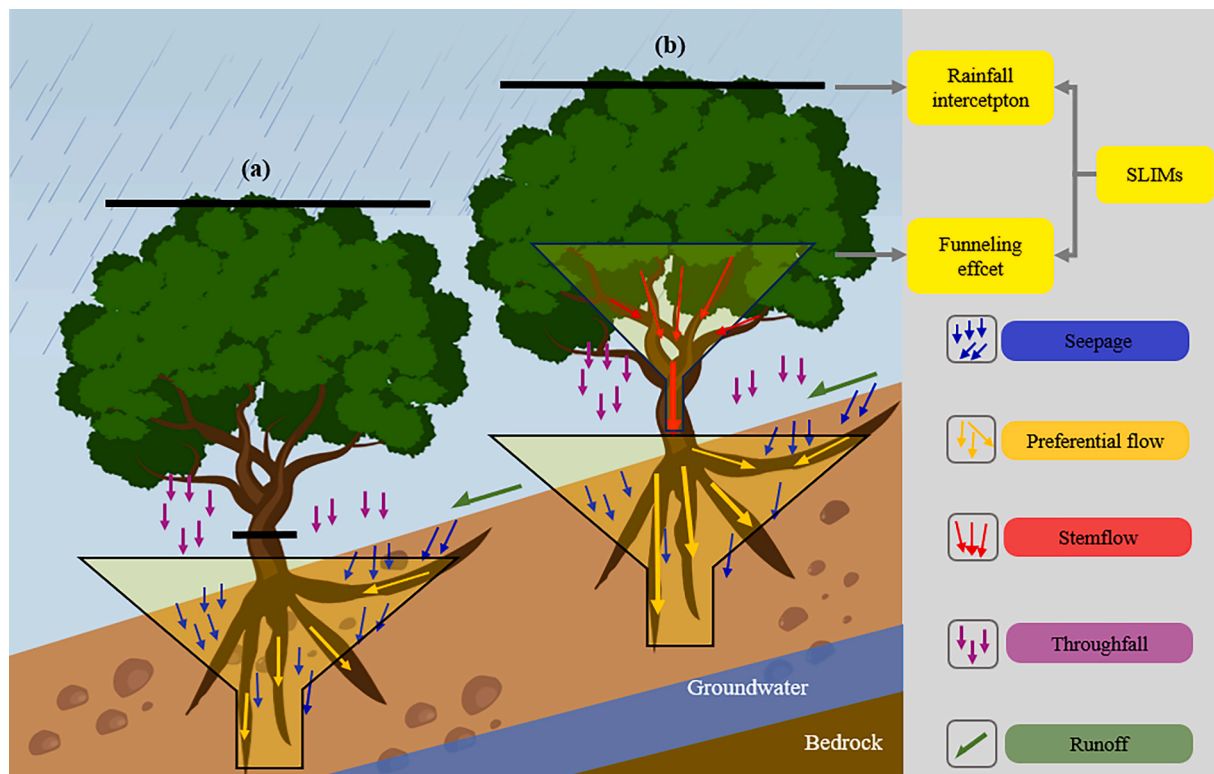


Fig. 4. Aboveground hydrological processes influenced by SLIMs, which encompass rainfall interception and funneling effect. (a) With no stem flow conditions, the capacity of vegetation to induce changes in soil moisture distribution diminishes. (b) Under the influence of vegetation’s rainfall interception and funneling effects, stem flow and throughfall are generated, altering both runoff and infiltration.

resulting in spatial variations in rainfall and slope surface moisture distribution during rain events. In addition, due to the presence of root systems, a significant amount of potential preferential flow is concentrated in the soil below the tree base, allowing throughfall to flow along the roots toward the center.

• Rainfall interception

Rainfall interception refers to the process by which all of the rainfall that falls on the vegetation surface is subsequently redistributed (Muzylo et al., 2009). Interception loss decreases with increasing rainfall intensity and cumulative rainfall, with higher interception loss observed during lighter rainfall events. During intense rainfall events, woody plants can intercept rainfall in the range of 0–5 mm (Keim and Skaugset, 2003). Research has also shown that annual canopy interception loss can range from 6 % to 45 % depending on factors such as tree species and stand density (Carlyle-Moses and Gash, 2011). Results from the modified TRIGRS model suggest that interception loss does not significantly affect the total amount of precipitation reaching the ground, but does alter the distribution of precipitation intensity and duration (Kim et al., 2013). Interception loss partially reduces runoff within the vegetation-affected area. Herbaceous vegetation, with lacks tall stems and leaves and has lower air conductivity, shows insignificant interception loss, throughfall, and stemflow. However, the stems and leaves of herbaceous plants increase the roughness of the slope, reducing the rate of surface runoff and increasing the rate of soil infiltration (Löbmann et al., 2020).

• Funneling effect

Johnson and Lehmann (2006) referred to the heterogeneity in precipitation water, slope surface water, and soil moisture caused by throughfall and stemflow as the double-funneling effect of trees, as shown in Fig. 4(b). Because it is difficult to quantify stemflow using conventional rainfall units (mm), most studies on forest rainfall interception focus on throughfall (Hanchi and Rapp, 1997). Aboal et al. (1999) found that although overland flow accounted for only 6.85 % of total precipitation, the concentrated rainfall in the tree infiltration area due to overland flow could be 12.8 times higher. Liang et al. (2011) observed the dynamic distribution of soil moisture over two years and used different dye tracers to track the infiltration pathways of throughfall and stemflow. The results show that soil moisture responds locally and rapidly to stemflow compared to without stemflow, suggesting that stemflow exacerbates the spatial and temporal heterogeneity of soil moisture, as shown in Fig. 4a and b. The funneling effect and rainfall interception together form leaf and shoot induced modifications (LSIMs) that affect stemflow, runoff, and throughfall.

2.2. Parameters of vegetation-induced modifications in hydrological processes

There are many factors that can influence VIMs in hydrological processes, and many parameters that can quantify the process to some extent. However, for specific form of hydrological process, some parameters or influences are not relevant. To illustrate, the rate of transpiration exerts a direct influence on intraplant hydrological processes. However, it is not feasible to quantify the impact of root channel on underground hydrological processes. This section tries to categorize parameters that have an impact on the efficiency of consumption and release, RIMs and SLIMs. And the parameters could be classified as vegetation specific, soil specific and environmental factors as shown in Table 1. The extent to which VIMs in hydrological processes are influenced by vegetation characteristics, soil characteristics and the environment. Parameters related to vegetation characteristics can be broadly categorized as vegetation type, biomass, and parameters describing roots, stems, and leaves.

2.2.1. Parameters related to vegetation properties

The VIMs differ between woody and herbaceous vegetation. The modification of hydrological processes by herbaceous vegetation is usually not carried out by a single individual. Dense herbaceous vegetation: 1) alter surface runoff velocity through leaf and stem cover (Gray, 1995; Morgan, 2007), 2) reduce soil water content through transpiration (Leung et al., 2015; Osman and Barakbah, 2006; Saifuddin and Osman, 2014), and 3) increases shallow surface hydraulic conductivity on slopes (Ghestem et al., 2011; Scanlan, 2009). Löbmann (2020) indicated that densely distributed herbaceous vegetation may have a higher evapotranspiration potential than woody vegetation at the catchment scale (Löbmann et al., 2020), although the short root systems of herbaceous vegetation are not yet able to influence soil suction deeper in the slope. Compared to herbaceous vegetation, woody vegetation has a relatively large individual structure that can alter the hydrological process of slopes through roots, stems, and leaves (Francini et al., 2021). Some studies have shown that woody plants are able to extract water from relatively deep soil layers (Schenk and Jackson, 2002), thus influencing soil moisture within a radius of three times the canopy diameter (Gray and Sotir, 1996) and creating spatial heterogeneity within a certain range.

Different species or the same species of vegetation, although showing similar patterns, may have different levels of impact due to differences in biomass, roots, stems, leaves, and morphology. Biomass, as an important parameter of vegetation condition, can to some extent reflect the uptake and consumption of substances by vegetation. However, biomass alone does not allow us to discriminate the degree of several inducing patterns. For example, high biomass may absorb more water and have stronger transpiration suction, but it also leads to a more significant local pore pressure increase and soil saturation due to stem flow. In addition, biomass may not have a significant effect on a particular hydrological process. Vegetation root biomass is proportional to root water uptake (Jackson et al., 1997; Schaffer et al., 2015), but this phenomenon is not only due to biomass accumulation. It is also related to the expansion of woody cellulose volume with increasing water content (Bordoloi et al., 2017).

Roots, as structures in direct contact with the soil, are the most studied parameters in slope stability research. Parameters related to root concentration in the soil include root density (RD), root length density (RLD), and root depth. RD is defined as the dry weight of roots per unit soil volume (Löbmann et al., 2020), and is a relatively easy root parameter to obtain (Gray, 1995). However, RD cannot reflect the structure, diameter distribution, and spatial distribution of roots within the measurement area, leading to conflicting conclusions when some researchers examine the relationship between RD and hydraulic conductivity or soil water retention curve. RLD is defined as the length of roots per unit soil volume or per unit soil surface area, and the rate of water extraction obtained from a soil layer depends on the RLD in specific soil regions and the absorption rate per unit root length (Stokes et al., 2009). Compared to RD, RLD better reflects the quantity and structure of roots. Root depth is more commonly discussed in the context of mechanical reinforcement (Mickovski and van Beek, 2009), but in recent years there has been an increase in research on root-induced the soil pore alteration and preferential flow through root channels, highlighting the importance of root depth in soil hydrological pathways (e.g., Ng et al., 2014; Schenk and Jackson, 2002; Zhu et al., 2019). In addition, the type of root system also has a significant effect. Fine and fibrous roots tend to clog existing macropores and reduce the soil hydraulic conductivity (Zhu et al., 2019). On the other hand, taproots and coarse roots often penetrate deeper soil layers and form root channels that communicate with the surface. Stemflow through these root channels to reach the deeper soil layers, forming preferential flow. Some studies have found correlations between the fine-coarse root ratio parameter and hydraulic conductivity, which may reflect the degree of root system obstruction and opening of soil hydraulic pathways (Haling et al., 2013; Zhu et al., 2019).

Table 1

A summary of the parameters associated with each VIM (consumption, release, RIMs, and LSIMs). Blank areas indicate that they have not been investigated yet. It remains uncertain whether or to what extent these areas are affected and thus require further attention (Aziz et al., 2017, Carminati et al., 2016, Chen et al., 2019, Della Vecchia et al., 2015), Fleischbein et al., 2005, Franzluebbers, 2002, Gared et al., 2015, Groeneveld et al., 1984, Guo et al., 2017, Hakansson (1990, Herwitz (1985, Jarvis et al., 2017, Linton and Nobel, 2001, Lynch, 1995, Marcacci et al., 2022, Murray, 2014, Ng et al., 2019c, Sandin et al., 2017, Simms and Yanful, 2001, Villordon et al., 2014, Warren et al., 2015, Yan and Zhang, 2015, Zhu et al., 2018).

Parameters	VIMs	Vegetation functions		Vegetation structures	
		Consumption and Release	RIMs	SLIMs	
Plant Root	Root area index (RAI)	(Gared et al., 2015; Garg et al., 2015; C. W. W. Ng, Z. J. Wang, et al., 2019)			
	Root length density (RLD)	(Aziz et al., 2017; Boldrin et al., 2017; C. W. W. Ng, R. Tasnim, et al., 2019; C. W. W. Ng, Z. J. Wang, et al., 2019)			
	Root architecture/ orientation	(Garg et al., 2015; Ng, Kamchoom, et al., 2016; Ng et al., 2015; Zhu et al., 2018)	(Ghestem et al., 2011; Lynch, 1995; Villordon et al., 2014)		
	Rooting potential/ability	(Chen et al., 2022; Colombi et al., 2018)	(Chen et al., 2022; Colombi et al., 2018; Groeneveld et al., 1984)		
	Fine-coarse root ratio		(Haling et al., 2013; Zhu et al., 2019)		
	Root density (RD)/Root Biomass/Root depth	(Garg et al., 2015; Schaffer et al., 2015; Zhu et al., 2018)	(Schaffer et al., 2015)		
Shoot	Xylem/Lignin	(Bordoloi et al., 2017; Brodribb & Cochard, 2009; Linton & Nobel, 2001)			
	Plant height	(Bordoloi & Ng, 2020; Gonzalez-Ollauri & Mickovski, 2017a)			(Bordoloi & Ng, 2020; Gonzalez-Ollauri & Mickovski, 2017a; Guo et al., 2017)
	Diameter at breast height (DBH)				(Gonzalez-Ollauri & Mickovski, 2017a; Guo et al., 2017)
	Shoot architecture				(Yuan et al., 2016)
	Basal diameter				
	Leaf area index (LAI)	(Gared et al., 2015; Garg et al., 2015; C. W. W. Ng, R. Tasnim, et al., 2019; C. W. W. Ng, Z. J. Wang, et al., 2019)			(Siles et al., 2010)
Leaf	Specific leaf area (SLA)	(Boldrin et al., 2017; Bordoloi & Ng, 2020)			
	Projected canopy area Or Crown width				(Gonzalez-Ollauri & Mickovski, 2017a; Guo et al., 2017; Herwitz, 1985; Liang et al., 2011)
	Vegetation coverage fraction (VCF)				(Chen et al., 2019; Fleischbein et al., 2005; Murray, 2014)
	Stomata conductance (SC)	(Boldrin et al., 2017; Brodribb & Cochard, 2009; Wong et al., 1979)			
	Transpiration efficiency (ET)	(Gared et al., 2015; Garg et al., 2015)			
	Aggregates size/ distribution /stability		(Bronick & Lal, 2005; Franzluebbers, 2002)		
Structure	Bulk density (BD)/degree of compactness (DC)	(Ng et al., 2014; Yan & Zhang, 2015)	(da Silva et al., 1997; Dal Ferro et al., 2014; Hakansson, 1990; Ng et al., 2014; Yan & Zhang, 2015)		
	Connectivity of the pore network		(Dal Ferro et al., 2014; Jarvis et al., 2017; Sandin et al., 2017; Zhu et al., 2019)		
	Pore size distribution (PSD)		(Bacq-Labreuil et al., 2018; Della Vecchia et al., 2015; Leung et al., 2023; Marcacci et al., 2022; Scholl et al., 2014)		
Soil	Soil type		(Haling et al., 2013)		
	Texture				
Texture	Soil water content (SWC)	(Bordoloi et al., 2017; Carminati et al., 2016; Warren et al., 2015)	(Bordoloi et al., 2017; Carminati et al., 2016; Della Vecchia et al., 2015; Haling et al., 2013; Simms & Yanful, 2001; Warren et al., 2015)		

Vegetation with similar RLD, RD, or fine-coarse ratio may still have different hydraulic properties at the slope scale. This is because the structure of roots (e.g., geometric shape, orientation) determines the spatial distribution of VIMs in the soil, exacerbating soil anisotropy. For example, some plants have lateral roots that grow mainly horizontally, improving horizontal pore connectivity and reducing pore pressure in deep soil (Zhu et al., 2019). The distribution and magnitude of transpiration-induced suction are influenced by the geometric shape of the root system (Ng et al., 2015; Yuan and Lu, 2005; Zhu and Zhang, 2015). Ng et al. (2016a) used a centrifuge model to test three types of artificial root systems: taproot, heart-shaped, and plant-shaped. The results showed that the soil surrounding the heart-shaped root had higher suction. This method, using artificial root systems, minimizes the soil suction caused by the roots and provides a better quantification of the suction caused by transpiration. Similarly, Leung et al. (2015) found that a heart-shaped root system can generate higher suction at a certain depth and radius range.

Quantification of stem and leaf parameters focuses primarily on vegetation consumption, release, and LSIMs. Stomata are the main channels through which plant leaves exchange gases with the external environment, and the degree of stomatal opening is a key factor influencing vegetation function, often represented by stomatal conductance (SC) (Wong et al., 1979). Specific leaf area (SLA) is defined as the one-sided area of fresh leaves divided by their dry mass. Boldrin et al. (2017) found a strong correlation between SLA, SC, and TE, suggesting that high SC may be a key factor that compensates for or causes low biomass (e.g., SLA). Rainfall interception consumes a portion of rainfall, delaying the fall of some raindrops and causing redirection, generating throughfall and stemflow. The overall structure of the canopy as well as stem and leaf characteristics, significantly influence this process (Yuan et al., 2016). Leaf area, stem water storage, the multilayered nature of the canopy, and evaporation during rainfall control interception losses. Basal diameter, as an important parameter of branch structure, can derive various stem and leaf-related parameters (Yuan et al., 2016), such as stem and branch bark surface area (Levia et al., 2015), leaf area index, leaf and stem biomass (Siles et al., 2010), and these indices affect stemflow productivity (SFP). Yuan et al. (2016) found that parameters that may influence stemflow in shrubs include basal diameter, leaf biomass, and larger branch angles. Under similar stem biomass conditions, dense, small-diameter branches have higher stemflow productivity than sparse, large-diameter branches.

2.2.2. Parameters related to soil properties

Soil characteristics such as soil type, density, and structure have a significant impact on the soil water characteristic curve (Cresswell et al., 1992; Gallipoli et al., 2003; Ng and Pang, 2000; Tarantino, 2009; Zhou et al., 2012). Parameters related to soil characteristics have been broadly categorized into soil structure and soil texture. Soil structure is defined as the spatial arrangement of the soil skeleton and pores at various scales (Rabot et al., 2018), without considering the chemical heterogeneity of gases and liquids in the soil skeleton and pores. There is an interactive feedback mechanism between the soil and the root system. Soil structure not only affects its own water retention and infiltration capacity but also affects the ability of roots to absorb water and nutrients, thereby affecting plant growth and the root system's infiltration process (Pardo et al., 2000). As a result, the root biomass, structure, and spatial distribution of the same vegetation can vary widely. These vegetation characteristics, in turn, alter soil structure and exacerbate the heterogeneity of soil hydraulic properties. Bulk density (BD), defined as the ratio of solid mass to undisturbed soil volume, is one of the most prominent indicators of soil structure (Rabot et al., 2018). BD is used to estimate soil compaction. In general, higher BD indicates greater compaction and lower permeability of the bare soil. However, BD is not considered a good indicator of soil functionality because it does not take into account important soil structures and types (Rabot et al., 2018). In order to eliminate the influence of soil type, the concept of "relative

compactness" (RC) is introduced, which better represents the compacted state of different soil types (da Silva et al., 1997; Håkansson and Lipiec, 2000). RC is defined as the ratio of the bulk density (BD) of the compacted soil to the reference bulk density (BD_{ref}) of the same soil. Excessive compaction reduces soil permeability but also affects root growth (Dal Ferro et al., 2014). Therefore, it is necessary to determine the optimal relative compactness (RC), which is defined as the ratio of the dry density of the compacted soil that meets ecological and engineering requirements to the maximum dry density of the soil (Ng et al., 2014). Ng et al. (2014) experimental model showed that high relative compaction affects root growth, with the root zone exhibiting high suction and delayed infiltration during rainfall events, but the influence of suction is shallower. Changes in BD, DC and RC may reflect changes in the overall hydraulic properties of the soil due to vegetation. However, they are difficult to characterize the differences in internal hydraulic properties of the soil, such as preferential flow and hydraulic conductivity of the matrix.

Aggregates are secondary particles formed by the combination of mineral particles with organic and inorganic substances (Bronick and Lal, 2005) and are fundamental elements of soil structure (Xiao et al., 2024). The size, distribution, and stability of aggregates affect the permeation process of the root system. Roots, especially fine or fibrous roots, tend to bypass the aggregates and fill the existing pores. Microaggregates bond together to form macroaggregates, and the bonds within microaggregates are stronger than those between microaggregates (Edwards and Bremner, 1967; Totsche et al., 2018), making connections between microaggregates a secondary choice for root penetration pathways. The pore size distribution (PSD) represents the proportion of different pore sizes (Bacq-Labreuil et al., 2018). Many studies have shown that vegetation has the ability to modify soil connectivity and expand the range of pore diameter distribution (Bacq-Labreuil et al., 2018; Bodner et al., 2014; Leung et al., 2023; Lu et al., 2020; Scanlan, 2009; Scholl et al., 2014), primarily through the tendency of roots to aggregate in large interconnected pores (e.g., channels that were initially empty or occupied by decomposed roots and later occupied by new roots) (Pierret et al., 1999; White and Kirkegaard, 2010). This process leads to more complex hydraulic properties of the soil (Bacq-Labreuil et al., 2018).

Soil texture, including mineral composition, water retention capacity, and particle size, varies widely among soil types. In general, clay soils have more organic matter and water need for plant growth and higher water retention capacity, making them more drought resistant. However, in clay soils, excessive transpiration from vegetation can cause dry cracking (Cheng et al., 2020), which can result in preferential flow that drains a significant amount of water, making it difficult for the inherently low permeability clay to replenish with water, thereby inhibiting root growth. Under extreme rainfall conditions, vegetation in clay soils is also more susceptible to oxygen deficiency, which limits root development. In sandy soils with limited organic matter and water, vegetation tends to allocate resources to primary root growth, limiting lateral root aggregation (Stokes et al., 2009). Nutrient-rich areas can promote the aggregation of lateral roots or fine roots (Hodge, 2004). The effect of root penetration on soil structure is also dependent on soil water content (Rabot et al., 2018). Physical constraints on root growth can be thought of in terms of the Least Limiting Water Range (Bengough et al., 2006; Dasilva et al., 1994), where root growth is restricted beyond a critical threshold. In arid or nutrient-competitive areas, roots often penetrate deeper to access water (Haling et al., 2013). Environmental factors such as climate, topography, and spatial distribution of vegetation influence both vegetation and soil properties, indirectly influencing VIMs in hydrological processes.

The functions and structures of vegetation play a crucial role in those hydrological processes discussed above, and their efficiency is controlled by several parameters (Gonzalez-Ollauri and Mickovski, 2017a; Leung et al., 2015; Ng et al., 2019b; Ng et al., 2022; Scholl et al., 2014). The authors have compiled a framework of each VIM for different

hydrological processes, as well as the parameters controlling in each VIM, as shown in Fig. 5.

3. Dynamic effects of soil strength

The VIMs in hydrological processes were discussed in Chapter 2 and summarized as consumption (photosynthesis and respiration), release (Transpiration), RIMs (root exudates, root channels, squeeze and connectivity), and LSIMs (rainfall interception and funneling effect). These VIMs control the magnitude of hydrological processes and exhibit strong temporal dependence (Arnone et al., 2016; Chirico et al., 2013). Each VIM may influence one or more hydrological processes. To understand the full extent of vegetation's hydrological influence on slope stability, it's important to first understand the positive and negative effects of different triggering mechanisms on soil strength at different rainfall stages. Soil strength refers to the soil's resistance to shear and is typically evaluated using Coulomb's law (Gonzalez-Ollauri and Mickovski, 2017b), which represents the maximum achievable state of soil stress through a graphical line called the "failure envelope" (Head, 1980). Soil shear strength varies with the moisture content in the pore spaces (Vanapalli et al., 1996), and has different characteristics in the unsaturated and saturated phases. Therefore, the soil condition during a rainfall event could be categorized into five phases: I) matric suction loss, II) pore water pressure development, III) pore water pressure dissipation, IV) matric suction recovery, and V) stabilization. This classification follows the time sequence to examine the positive or negative effects of these VIMs on soil strength at the local scale, as shown in Fig. 6.

Vegetation functions, particularly transpiration, are generally considered to be the primary mechanism by which vegetation enhances soil strength and improves slope stability via hydrological processes (e. g., Bordoni et al., 2024; Liu et al., 2022; Ng et al., 2019a). However, transpiration is highly sensitive to soil moisture and environmental conditions (Dhakal and Sidle, 2004; Sidle and Ochiai, 2006). During wet events, transpiration efficiency decreases (Aparecido et al., 2016; Hadiwijaya et al., 2020; Yasutake et al., 2018), and the contribution of vegetation-induced soil suction decreases (Leung et al., 2015). After the initial transpiration, the vicinity of roots is typically in an unsaturated state, maintaining higher suction before the onset of rainfall (Mahmood et al., 2013; Voytek et al., 2019). As rainfall occurs, the soil gradually moves toward saturation (Phase I), and suction decrease rapidly during this phase, resulting in the loss of additional cohesion provided by suction. As the soil enters saturation (Phase II), transpiration efficiency decreases. Although it still extracts some water from the soil to the atmosphere, soil saturation increases hydraulic conductivity, resulting in an infiltration rate greater than the drainage rate. The pore water pressure gradually increases and stabilizes, while the effective stress decreases and then stabilizes. Both of these stages result in a decrease in soil shear strength, with the loss of suction causing a rapid decline. Under the combined effect of transpiration and gravitational drainage, pore water pressure gradually dissipates (Phase III), and transpiration efficiency gradually improves, leading to an increase in effective stress (Chadwick et al., 2022). After the transition from saturation to unsaturation (Phase IV), suction gradually begins to recover, and additional cohesion reappears, eventually reaching a stable state (Phase V). Soil shear strength shows a gradual recovery trend during the last three phases. Overall, vegetation functions (transpiration) have a positive effect on slope stability, even under extreme rainfall conditions, as shown in Fig. 6a and g.

Growing roots can exert pressure on the surrounding soil (Chen et al., 2022; Colombi et al., 2018). Compacted soil in the unsaturated state has a higher suction than before compaction and a lower permeability coefficient and pore water pressure in the saturated state. Therefore, during rainfall and the subsequent drying phase, roots have a positive effect on slope stability, as shown in Fig. 6(c). However, this positive effect may not be significant under extreme rainfall conditions. The soil

may also crack due to compaction (Cheng et al., 2020). Cracked soil resulting from root-induced compaction has lower suction and higher permeability coefficients than before cracking, resulting in a negative effect on slope stability. This negative effect is most pronounced during periods of high cumulative rainfall, as shown in Fig. 6(b).

A discussion of how root secretions increase aggregate stability was presented in Section 2.1. Increased aggregate stability results in an increased ability of the soil to maintain its structure under various stress conditions (Frei et al., 2003). Increased aggregate stability allows the soil to maintain its structure under varying stress conditions. Even under high moisture conditions, the soil structure is less susceptible to strain failure due to the decrease in cohesion and effective stress. Root exudates reduce differential deformation within the slope, resulting in a positive effect on local soil stability, as shown in Fig. 6(d).

Ghestem et al. (2011) summarized three types of root channels that can modify or create preferential flow paths (discussed in Section 2.1) and discussed how root architecture influences preferential flow paths under static conditions. Based on this, under dynamic conditions, root channels with different hydrological characteristics can be classified as follows: channels occupied by living roots that tend to accumulate pore water pressure, channels occupied by living roots that tend to dissipate pore water pressure, old channels blocked by roots, and decaying root channels. Channels occupied by living roots with dead ends tend to accumulate pore water pressure, resulting in an overall negative effect on slope stability, as shown in Fig. 6(h). The presence of living roots creates preferential flow pathways, causing a reduction or even disappearance of matric suction in the soil matrix. This phenomenon has a negative effect on soil strength even during the antecedent stage of rainfall (Chen et al., 2021; Li et al., 2023; Luo et al., 2019). When rainfall occurs, water rapidly infiltrates into the root channels, resulting in positive pore water pressure. The high hydraulic gradient formed in the root channels facilitates water infiltration into the surrounding soil with relatively low hydraulic gradients, which accelerates the loss of matric suction and the reduction of effective stress in the surrounding soil, exacerbating the negative effect. However, due to the water absorption and swelling of the roots, as shown in Fig. 2a and b, the water flow in these preferential flow channels is reduced or even cut off, limiting the development of pore water pressure and lateral infiltration to some extent, resulting in a slower increase in the rate of negative effect.

After rainfall, the vegetation first drains the water in the root channels, gradually dissipating the pore water pressure and returning to a stable state where the negative effect is primarily caused by the loss of matric suction. Channels occupied by living roots but with connectivity tend to dissipate pore water pressure and have a positive effect on slope stability during periods of declining slope stability (due to cumulative rainfall) as shown in Fig. 6(i). Although these root channels also have a negative effect during the early stages of rainfall, they do not create high hydraulic gradients during rainfall events, resulting in minimal impact on the surrounding soil. In the later stages of pore water pressure development, they may even serve as drainage pathways, slowing the rate of decrease of effective stress in the surrounding soil. Old channels blocked by roots have a positive effect on slope stability during rainfall events, as shown in Fig. 6(j). Studies have shown that roots tend to aggregate into large pores (Stewart et al., 1999; White and Kirkegaard, 2010), blocking the original preferential flow paths, especially fine or fibrous roots. Dense fine roots and root hairs create additional suction in the large pores through capillary action, providing pre-event hydrological reinforcement. After the loss of matric suction in the later stages of Phase I, matric suction is primarily induced by the root pattern rather than transpiration (Leung et al., 2015). Research has shown that root-clogged pores may have lower permeability than the soil matrix (Ghestem et al., 2011), and may still have a positive effect on slope stability. Channels blocked by decaying roots have similar negative effects as channels occupied by living roots, which tend to accumulate pore water pressure. However, these channels do not have living roots capable of absorbing and swelling water and therefore cannot suppress

the development of pore water pressure and lateral infiltration within the root channels. This results in a continuous increase of the negative effect, as shown in Fig. 6(k).

The stem and leaf induction patterns (double funneling effect and interception or delay) create aboveground hydrological processes that do not exist in bare soil (throughfall and stemflow), which affect the infiltration boundary conditions of underground hydrological processes (infiltration and preferential flow), including total infiltration, time of rainfall reaching the slope, and location of water collection (Stokes et al., 2014). The interception or delay effect consumes a portion of the rainfall and delays its infiltration, indirectly providing a positive effect during the early stages of rainfall. As the rainfall progresses, the canopy gradually become saturated, and the positive effect diminishes. After rainfall ceases, the remaining water in the canopy provides infiltration supply for a period of time, resulting in a negative effect, as shown in Fig. 6(e). During extreme rainfall events, the interception volume decreases (Gonzalez-Ollauri and Mickovski, 2017a; van Dijk and Bruijn-zeel, 2001), resulting in a shorter lag time. As rainfall accumulates, the water intercepted by the canopy continuously converges towards the

tree trunk and flows down to the base, forming an aboveground hydrological process resembling a funneling. The area beneath the base of the tree is where root channels are most densely distributed, and the stemflow quickly forms preferential flow within these channels (Kobayashi et al., 2000). The soil matrix surrounding the root channels rapidly loses its matric suction and enters saturation prematurely. As the pore water pressure dissipates, this negative effect disappears. The response of preferential flow to the funneling effect becomes more rapid with increasing rainfall intensity, as shown in Fig. 6(f).

The trend changes of the positive and negative effects of the above VIMs on the soil strength, the degree of influence and the nodes of rate change of the degree of influence are represented qualitatively in Fig. 6, and they are influenced by different vegetation characteristics, soil characteristics and parameters of the antecedent condition, which were discussed in Section 2.2 and tried to be summarized in Table 1.

4. Dynamic effects of slope stability

In a broad sense, “landslide” refers to the phenomenon of material

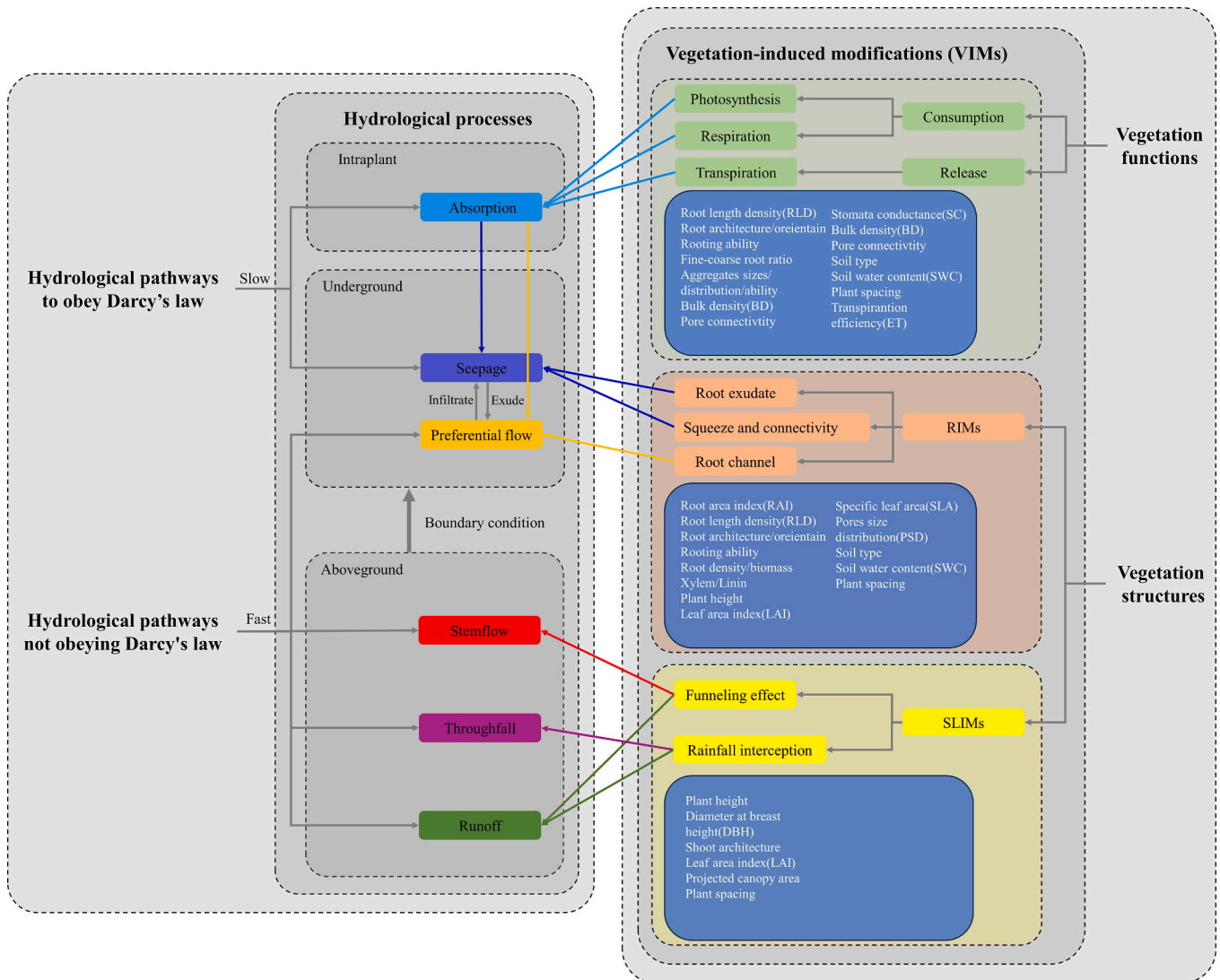


Fig. 5. A systematic framework for describing the intrinsic relationships between VIMs and hydrological processes. It includes section 2.1, which examines the specific relationships between each VIM and hydrological processes, and section 2.2, which examines the parameters that influence the impact of each VIM on specific hydrological processes. The VIMs and their associated influencing parameters are shown on the right, with lines connecting specific VIMs to the hydrological processes they influence. The conceptual framework of slope hydrological processes is outlined on the left. In addition, hydrological processes within and above vegetation are not only influenced by VIMs, but also affect underground hydrological processes. There is a feedback mechanism between Darcy flow in porous media and preferential flow in non-porous media in underground.

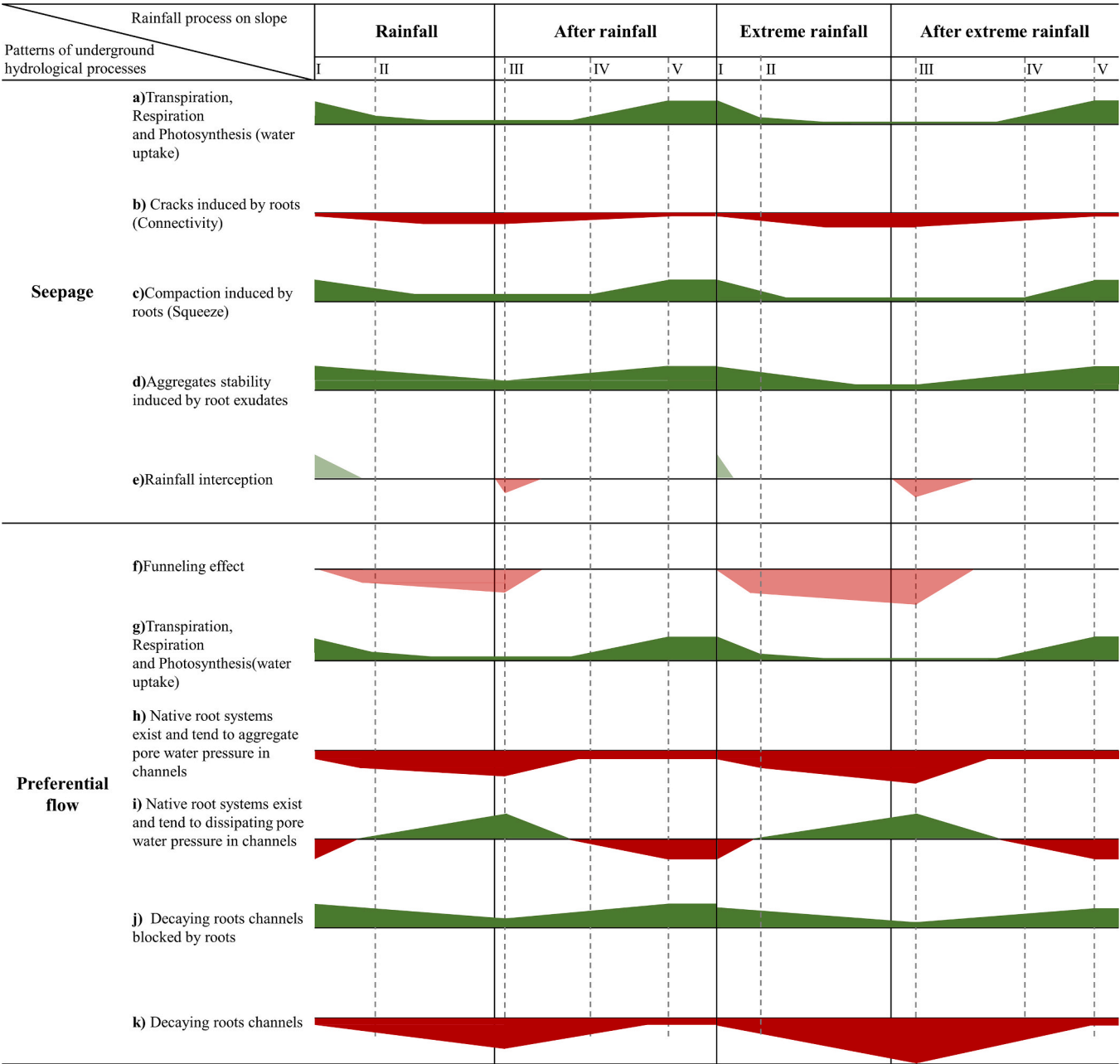


Fig. 6. Positive and negative effects of hydrological processes after vegetation induction on the strength of local soils at the margins and their extent changed with time. Where: (I) matric suction loss phase, (II) pore water pressure development phase, (III) pore water pressure dissipation phase, (IV) matric suction recovery phase, (V) stability phase. green indicates that the process has a positive effect on slope stability, red indicates that the process has a negative effect on slope stability. thickness indicates the degree of effect, and light color indicates that the effect of the process on slope stability is indirect. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(rock, soil, artificial fill, or a combination) forming a slope that moves downward and outward (Cruden and Varnes, 1996). Many studies consider mass movements such as rockfalls, soil erosion, or debris flows to be part of landslides in the broad sense, although “mass movements” may be a more accurate classification. Landslides, broadly defined, are important research subjects in various fields, including geotechnical engineering, soil science, plant science, ecology (Stokes et al., 2014). However, the interdisciplinary nature of this research often leads to conceptual ambiguities due to different perspectives on similar processes. Soil and plant science studies, due to their scale and objectives, often focus on VIMs in hydrological processes and their effects on matric suction or soil strength at fine scale. However, considering the reduction or loss of matric suction as the only or significant factor in landslide

occurrence (Sidle and Bogaard, 2016), and oversimplifies the understanding of slope scale mechanical mechanisms and other influencing factors and heterogeneity at the catchment scale (e.g., Boldrin et al., 2017; Hawke and McConchie, 2011; Liu et al., 2022; Osman and Barakbah, 2006; Reubens et al., 2007; Wright et al., 2007). On the other hand, geotechnical and ecological studies mainly focus on slope stability mechanisms and regional influencing factors, overlooking the influence of vegetation structure on initial infiltration and preferential flow, as well as the temporal effectiveness induced by vegetation functions (e.g., Bittelli et al., 2012; Godt et al., 2009; Li et al., 2021; Ng et al., 2021). Due to the transition from fine scale to slope scale and the complexity of landslide mechanisms, vegetation-induced hydrological processes exhibit a high degree of spatiotemporal heterogeneity in their effects on

slope stability. Therefore, to explain the dynamic response of slopes with different characteristics (e.g., slope gradient, soil depth, types, and bedrock properties) to vegetation-induced hydrological processes under rainfall events, it is necessary to study them in a time series and at the slope scale.

4.1. From soil strength to slope stability

The effects of different VIMs on soil structure and strength are typically significant at fine scale. Due to scale effect, the hydrological effects of vegetation on soil strength do not necessarily represent the effects on slope stability. On the fine scale, the root parameters exert a certain influence on the soil's vertical hydraulic conductivity, cohesion and friction angle. However, it is not possible to take into account the relationship between the distribution of the root system over a certain space and the hydraulic diffusion coefficient, and the lateral hydraulic conductivity is often neglected. Moreover, slopes represent the fundamental unit of landslide occurrence, exhibiting their own distinctive stress state and properties (Debieche et al., 2012; Sidle and Bogaard, 2016). Recent studies have demonstrated that the presence of vegetation significantly increases the hydraulic conductivity coefficient of shallow soil with dense root distribution, relative to that of the underlying soil layer. Additionally, the slip surface of slope instability frequently occurs at the lower interface with sparse root distribution (Hu et al., 2024; Lin et al., 2024; Qin et al., 2022). This is due to the fact that at the onset of precipitation, the soil is in a non-saturated state, and vertical infiltration represents the primary mode of water transport. However, the "bottle-neck effect" causes the rapid infiltration of rainwater to saturate the interface between the two with great rapidity. This results in intensified lateral infiltration and the generation of permeability simultaneously, while the pore water pressure rises rapidly. It is possible that the hydrological effect of vegetation on soil strength may be overestimated when scaling up to the slope scale, and that the weakening stage of the hydrological effect of vegetation may be underestimated (Guo et al., 2020b; Lin et al., 2024). Before discussing scale of slope, it is necessary to clarify the stability and hydrological effects of slopes.

The term "landslide" in a narrow sense refers to the phenomenon whereby a portion of a slope undergoes shear movement along one or more surfaces within the slope under the influence of gravity (Murgia et al., 2022). The sliding motion is generally divided into two types: planar sliding and rotational sliding. The sliding surface is defined as the planar surface along which the mass of soil in the slope slides or separates (Morgenst and Price, 1965). The key indicator for quantifying slope failure (landslide occurrence) is the factor of safety (FS), which is the ratio of the resisting forces to the driving forces on the potential sliding surface. Typically, if the FS is less than 1, the slope is considered stable, and if it is greater than 1, the slope is failing. The resisting forces include the shear strength of the soil and roots (consisting of frictional and cohesive strength) if there are roots crossing the potential slip surface. Driving forces include the gravitational component of the landslide mass and overlying materials (such as vegetation) in the downslope direction, and the component of seepage forces generated by unstable seepage (if present) in the downslope direction. Due to the structural characteristics of the soil, different parts within the potential landslide mass experience differential deformations, which affect the stress state of the slope and the potential sliding surface (Lehmann and Or, 2012). Changes in slope hydrological conditions caused by rainfall are one of the most common triggering mechanisms for soil slope failures (Meusburger and Alewell, 2008; Osman and Barakbah, 2006). The hydrological response of slope stability includes infiltration response and the preferential flow response (Shao et al., 2018). The determination of soil hydrological regimes requires the establishment of the soil water characteristic curve (Vangenuchten, 1980), which shows that matric suction decreases with increasing saturation (Ng and Pang, 2000). As soil increases saturation within the slope, the cohesive strength induced by matric suction gradually decreases until it ceases to exist in the

saturation phase (Chen et al., 2020; Sattler et al., 2021). The increase in pore water pressure in the saturated state results in a decrease in effective stress and consequently a decrease in shear strength. This transition from unsaturated to saturated conditions, accompanied by increasing pore water pressure, undermines the soil strength of the slope (Kong et al., 2022). Slope stability is highly heterogeneous in response to preferential flow networks. Fractures and large pores associated with the slope surface form preferential flow networks that drive the response of the surrounding soil to infiltration and exacerbate the anisotropy of the slope's response to seepage, resulting in differential deformation of the slope. Blocked fractures or large pores (such as collapsed root channels) may be delayed or prevented from participating in the hydrologic process by the surrounding preferential flow network. Dead-end preferential flow paths crossing or near the potential slip surface can cause a sudden increase in pore water pressure, resulting in a decrease in shear strength. Continuous preferential flow pathways can also drain water from the interior of the slope, maintaining the shear strength of the potential slip surface. The increase in the slope's own weight due to rainfall and the seepage forces generated by unstable seepage increase the driving forces on the landslide mass. Therefore, the factor of safety (FS) is dynamically controlled by the hydrological conditions and response time at different positions within the slope. Furthermore, the highly dynamic FS needs to consider the influence of vegetation on slope stability, including the location and extent of vegetation influence, making the response of slope stability to hydrological processes more complex (Reubens et al., 2007; Zhang et al., 2022). The hydrological effects of vegetation on slope stability have been investigated and quantified in last decade (Bordoloi and Ng, 2020) and reviewed (e.g., Bordoloi and Ng, 2020; Liu et al., 2022; Löbmann et al., 2020; Ng et al., 2022; Stokes et al., 2009; Stokes et al., 2014). The main research methods can be broadly categorized as: laboratory model experiments, field combined with simulation experiments, and physical models. With the development of related instruments and driving devices, centrifuge models have been widely applied in landslide model experiments because they can simulate stress levels under realistic conditions in small-scale models (Fang et al., 2023). The centrifuge model experiments on rainfall-induced landslides under the influence of vegetation have investigated threshold changes and hydrological effects under both positive and negative pore water pressure (Leung et al., 2017; Ng et al., 2016a; Ng et al., 2017). The experimental results typically show that VIMs in suction alter the critical infiltrations of the FS, dynamically reflecting the correlation between VIMs in matric suction and slope stability. However, although the centrifuge machine can simulate slope scale features, hydrological processes derived from fine scale experiments are difficult to reflect the effects of vegetation spatial distribution and complex preferential flow paths and characteristics (Fan et al., 2020). In addition, hydraulic conductivity and infiltration parameters are highly scale dependent, and results at fine scale may not be satisfactory in slope scale (e.g., Brooks et al., 2004; Mohanty and Mousli, 2000; Sidle et al., 2007). Field combined with simulation experiments mainly interpret the experimental results through statistical correlation analysis or physically based model analysis (e.g., Garg et al., 2015; Leung and Ng, 2013; Li et al., 2021). Root length density (RLD) can be used as an indicator of slope stability (Osman and Barakbah, 2006). Qin et al. (2022) observed significant changes in hydraulic conductivity within the root influence zone, which negatively affected slope stability, and attributed this phenomenon to preferential flow generated by root channels. However, Singh et al. (2018) experiments showed the opposite phenomenon, attributing it to root occupation of pores and compacted soil. This is mainly because such studies have not mechanistically considered the inducing patterns of vegetation. Through the discussion in Chapter 2, we understand that, in fact, preferential flow and root occupation of pores or compacted soil coexist within the slope, and the positive and negative effects on FS, matric suction, permeability coefficient, and other indicators depend on the combination of various inducing patterns. Comparatively, Gonzalez-Ollauri and Mickovski

(2017a) derived the functional relationship between vegetation evaporation (drying process), throughfall, stemflow (wetting process), and matric suction based on a physical model, which resulted in dynamic changes of FS at different depths. However, this study still downplays the effects of preferential flow and lateral spatial heterogeneity. Slope stability models (SSMs) coupled with root systems, which have been widely studied in the fields of ecology, geology and geomorphology, and engineering, have been reviewed (Murgia et al., 2022). The hydrological and mechanical effects induced by root systems have gradually shifted from constant to variable, considering, for example, the varying degree of vegetation influence with slope changes, (e.g., Arnone et al., 2016; Cislighi et al., 2017; Cohen and Schwarz, 2017; Mao et al., 2014; van Zadelhoff et al., 2022). Under extreme rainfall conditions, the effect of vegetation on slope stability through canopy interception and evapotranspiration is weak compared to the root-influenced soil mechanical and hydrological properties (Kim et al., 2013; ZHANG et al., 2010), and root-induced hydrological effects are often ignored. Therefore, many physical models still do not systematically consider the influence of hydrological processes (Zhang et al., 2022).

Some studies considering the influence of vegetation on slope stability have started to focus on hydrological effects, but preferential flow and non-uniform infiltration conditions are often ignored (e.g., Feng et al., 2020; Liu et al., 2016; Ni et al., 2018), because the incorporated hydrological models still do not systematically consider VIMs. Conservative treatment of FS as a stability indicator at the slope scale may obscure the fine scale details of vegetation, making it difficult to integrate hydrological models that account for vegetation effects, which are highly heterogeneous, into slope stability models (SSMs). In order to reproduce the multi-process and multi-variable problem as accurately as possible, future slope stability modeling methods for vegetation hydrological effects must integrate VIMs at the fine scale and establish constitutive models of hydrological processes with anisotropy and time dependence. Based on these constitutive models, slope stability models should be developed to account for spatiotemporal effects (Stokes et al., 2014).

4.2. Spatiotemporal patterns of dynamic effects of slope stability

To better understand and demonstrate the mechanism of change, we introduce the space–time cube model of analysis. The use of a space–time cubes allows researchers to effectively visualize and analyze spatiotemporal characteristics within motion trajectory datasets in geovisualization (Wagner et al., 2020). Space-time cubes are found to provide a comprehensive understanding of soil hydrologic processes (Vereecken et al., 2022).

Using $y = f(x, y, z, p, t)$ is able to store attributes in a certain time and space scale, which x, y and z store spatial attributes, p stores multi-dimensional properties, and t represents time. When constructing a space–time cubes model covering dynamic vegetation hydrological effects, the spatial structure of slopes at the slope scale ($y = f(x, y, z, p_0, t_0)$, p_0, t_0 are constant), and the influence of multiple vegetation hydrological effects on soil strength at the fine scale ($y = f(x_0, y_0, z_0, p, t)$, x_0, y_0, z_0 are constant) need to be considered, which had been discussed in section 3. To facilitate further discussion, we use a typical event as an example to explore the extent and magnitude of vegetation hydrological effects on slope stability.

The Caiyuan Basin (26.72°–26.78°N, 117.89°–117.95°E), with an area of approximately 25 km², is situated in Fujian Province, a region characterized by high vegetation cover on the southeast coast of China, with a subtropical maritime monsoon climate. The area is characterized by an evergreen closed broadleaf forest dominated by *Cunninghamia lanceolata*, with a thick soil layer and a well-developed root system. Hydrological processes are strongly induced by vegetation. Furthermore, the occurrence of typhoons and heavy rainfall contribute to the formation of rainfall-triggered landslides throughout the year. The area

experienced a significant amount of precipitation from 14 to 23 June 2010, with a cumulative rainfall of 356 mm. The highest hourly rainfall was recorded at 2:00–3:00 on 14 June, reaching 22.5 mm/h, resulting in the generation of over 1,000 landslides, as shown in Fig. 7(a). As with previous landslides, the landslides generated by this event were typical rainfall-triggered shallow landslides, with small individual landslide areas and depths, occurring in clusters.

The space–time cubes model was employed to incorporate the effects of vegetation dynamics. This was achieved by: 1) obtaining the parameters affecting the aboveground hydrological processes of vegetation in the region, such as vegetation stem height, canopy cover, etc., through remote sensing. Obtaining the root parameters and hydrological parameters of the vegetation at different times and depths through field and indoor experiments. VIMs in hydrological processes occurring simultaneously and at the same location are incorporated as properties ($y = f(x_0, y_0, z_0, p, t_0)$) into a specific time slice and spatial cross-section of the space–time cubes, as illustrated in Fig. 7(d). 2) Each voxel within the slope is assigned a time series and associated properties, with the size of the voxel dependent on the spatial resolution of the study area, as illustrated in Fig. 7(c). The spatial resolution of the digital elevation model (DEM) in this study area is 5 m × 5 m. 3) The slope units within the study area should be delineated. In this study, the delineation of the study area was conducted through a combination of hydrological analysis and the utilization of the r.slopeunits (Alvioli et al., 2016), as illustrated in Fig. 7(b). The test and sampling points were distributed in a manner that ensured the parameters obtained were interpolated to the entire study area, serving as initial conditions for the calculation of hydrological processes across the entire study area. 4) The results of hydrological process calculations in the study area were used as the initial conditions for the stability calculation of the slope units, and FS of each slope evolved with time.

Excessive slopes displacement and the formation of continuous plastic zones are considered characteristics of slope failure (Zou et al., 2021). Studies have shown that both the stage of matric suction loss from unsaturated to saturated conditions during rainfall events and the continuous action of pore water pressure in the saturated state can lead to landslides (Fang et al., 2023; Lehmann and Or, 2012; Rahardjo et al., 1995; Tsai, 2011). During rainfall periods, soil strength within the slope decreases, leading to a gradual reduction in slope stability. However, the contribution of each location to slope failure varies. Soil strength above the potential slip surface is worth paying attention to, but the water content, time, and path through the potential slip surface are critical in determining the location and timing of landslides (Sidle and Bogaard, 2016). Particularly during high-intensity rainfall events, slope failures typically occur along shallow slip surfaces (Ciurleo et al., 2017). In addition, root channels serve as potential preferential flow paths, providing a rapid infiltration pathway for rainwater into deeper soil layers (Guo et al., 2020a). The high hydraulic conductivity differences at the root tip lead to a sudden increase in pore pressure, which to some extent controls the formation of the sliding surface (Qin et al., 2022). Therefore, when considering the positive and negative effects at the slope scale, it is necessary to assign importance indicators to the spatiotemporal cubes. To further discuss the consequential dynamics effects of slope stability, we discuss the changes in a slope scale cross-section over five phases of rainfall event, as shown in Fig. 8.

Regardless of the function or structure of vegetation, its influence on slope stability is primarily manifested within a certain range of root distribution. The areas within the influence range can be divided into the preferential flow zones with dense root systems and the seepage (soil matrix flow) zones with weaker root influence (Xing et al., 2023). During the matric suction loss phase, rainfall infiltrates rapidly along the tree base or surface cracks, and underground structures such as fractures, connected by roots also participate in preferential flow. The interconnected underground preferential flow network (preferential flow zones) becomes an additional infiltration boundary for the seepage

zones, and uneven infiltration begins (Li et al., 2023). Compared to outside the vegetation influence range, the initially accumulated high suction dissipates rapidly, leading to rapid saturation of the seepage zones near the roots. The positive effects caused by vegetation function weaken and the role of root channels gradually becomes dominant. The positive effects in the seepage zones may be maintained by aggregation due to compaction and secretion, while the tree base, dead ends, and potential preferential flow channels connected by roots exhibit negative effects. A wetting front is initially formed and develops rapidly near roots and surface cracks. Some preferential flow channels can cause rapid water flow in the vicinity of potential slip planes, resulting in a reduction in soil strength at potential slip plane areas, as shown in Fig. 8 (a).

During the pore water pressure development phase, preferential flow channels with dead ends maintain high pore water pressure. The preferential flow channels with high hydraulic heads affect the adjacent matrix areas, and the negative effect zones continue to expand (Fig. 8 (b)). The wetting front during this stage develops unevenly, with the front beneath the trees potentially reaching the slip planes. This can cause a significant reduction in soil strength at these potential slip planes, which is often the primary cause of rainfall-induced landslides (Wang et al., 2019).

During the pore water pressure recovery phase, the positive effects of vegetation are gradually restored. The slope soil gradually transitions from unsaturated to saturated and then back to unsaturated conditions from shallow to deep. As the slope enters the post-rain drying process, the surface soil is gradually desaturated under the influence of water gradients. Vegetation functions (transpiration) are gradually restored, accelerating the drying of the soil near the roots. The upper boundary of the wetting front shows a wavy pattern, with the trough position usually just below the trees (Fig. 8(c)). Under the influence of gravity and hydraulic gradients, the wetting front continues to advance into the deeper soil, resulting in a larger area of soil strength reduction at potential slip plane areas. Although vegetation promotes deep soil drainage during this phase, it takes longer, especially under extreme rainfall conditions (Marotti et al., 2023). For this reason, rainfall-induced landslides are most likely to occur during this phase.

During the matric suction recovery phase, the preferential flow zones with high permeability are the first to dry out. As vegetation continues to transpire, soil suction in the root zone gradually recovers. The negative effects caused by root action gradually diminish or become positive, and transpiration becomes the dominant factor again. At this phase, vegetation has an overall positive effect on slope stability (Fig. 8(d)). As saturation and suction stabilize at different positions, vegetation provides continuous additional cohesion to the soil within the root influence zone until the next rainfall event occurs (Fig. 8(e)). For the soil outside the influence zone, the lateral effects of vegetation increase or decrease the soil strength within the influence zone, thereby increasing or decreasing the deformation of the soil within the influence zone. The soil-vegetation coupling shows a certain stiffness difference compared to the soil in the unaffected area, which affects the overall deformation of the slope (Fan and Lai, 2014).

In summary, the structure and functionality of vegetation result in significant uneven wetting and drying cycles of slope soils. Vegetation structure plays a controlling role during the wet phase (I, II), while vegetation function controls the dry phase (III and IV) and early phase (V). The seepage zones exhibit slow infiltration and drainage of rainfall with a delayed effect on slope stability. Potential preferential flow channels respond rapidly to rainfall events, generating negative effects on slope stability, especially when high pore water pressure are present near potential slip planes. The recovery rate of preferential flow zones during the dry phase is faster than that of soil seepage zones.

In addition, the characteristics of the slope itself can in turn influence the growth and distribution of vegetation, thereby increasing or decreasing the hydrological effects of vegetation on slope stability. For example, high-angle slopes are particularly sensitive to the response of

rainfall self-weight and seepage force, where the hydrological effects of vegetation may not be the primary factor. Root shape and length influence soil water retention and depth, and the hydrological effects induced by shallow-rooted plants during prolonged rainfall have minimal effects on slope stability (Forbes et al., 2013; Liu et al., 2016), as shown in Fig. 9. The thickness of the soil cover is a particularly important piece of information (Zhang et al., 2022). Compared to deep soils, vegetation is more likely to detect and grow into the bedrock beneath in shallow soil layers (Schenk, 2008). More roots passing through or at potential slip planes will enhance the hydrological effects of vegetation on slope stability, as shown in Fig. 10.

5. Future perspectives

5.1. Gaps between models and the real soil hydrological processes

Soil hydrological processes determine the timing and location of water infiltration, flow, and redistribution in the soil (Seneviratne et al., 2010). Understanding the dynamic soil hydrological processes is critical for establishing hydrological models at various scales related to slope stability, infiltration, analysis, and mitigation of rainfall-induced landslides (Novick et al., 2022; Vereecken et al., 2022; Wang et al. 2020a, Wang et al., 2020b). Hydrological models at various scales require reliable support from dynamic soil hydrology. Early research focused primarily on establishing quantitative models for infiltration, (e.g., Horton, 1940; Parlange, 1971). However, these classical infiltration models often assume that groundwater processes follow the Darcy's law for homogeneous porous media. This assumption neglects the existence of preferential flow paths and cannot predict specific non-uniform infiltration and non-Darcy flow (Fan et al., 2020). At the scale of slopes, due to scale effects, this deficiency can be calibrated through correction factors to achieve the desired overall infiltration rate. However, for landslide studies, the timing and paths generated by preferential flow paths are particularly important for slope stability analysis. Under extreme rainfall conditions, slopes may experience earlier development of pore water pressure on potential slip surfaces due to the presence of preferential flow paths, significantly advancing the actual occurrence time of landslides relative to predictions. At regional and even global scales, parameters derived from PTFs often do not account for plant roots and macropores, resulting in lower overall soil saturated hydraulic conductivity than values that account for soil structure (Fatichi et al., 2020). The simulation of dynamic soil hydrological processes is less developed due to the lack of parameterization of transient soil structure and the high cost and time-consuming nature of measurements (Lu et al., 2020; Vereecken et al., 2010).

In recent years, there has been increasing attention to understanding and parameterizing soil hydrological processes under complex conditions, such as multiple soil layers and root effects (Blöschl et al., 2017; Saini et al., 2016). However, as discussed in section 2, soil hydrological processes exhibit high variability and uncertainty, and our ability to describe and predict dynamic hydrological processes lags behind our needs. Current models still struggle to reproduce observed dynamic hydrological processes in real soils (Assouline, 2013), and the existence of preferential flow casting doubt on the effectiveness of classical infiltration equations in heterogeneous soils (Guo and Lin, 2018). It is worth mentioning that since 1982, Beven and Germann (1982) has consistently advocated the need for more experimental data to support the incorporation of preferential flow concepts into hydrological models. The Darcy-Richards framework has recently been extended to incorporate preferential flow (Beven and Germann, 2013). These methods can be broadly categorized as the correction factor method, the dual-computational domain method, the dual-permeability method and the dual-porosity method (Gerke, 2006). However, these methods are contingent upon the Richards equation and its associated assumptions of equilibrium gradients and homogeneous soil properties at small scales. Furthermore, they have difficulty reproducing observations of

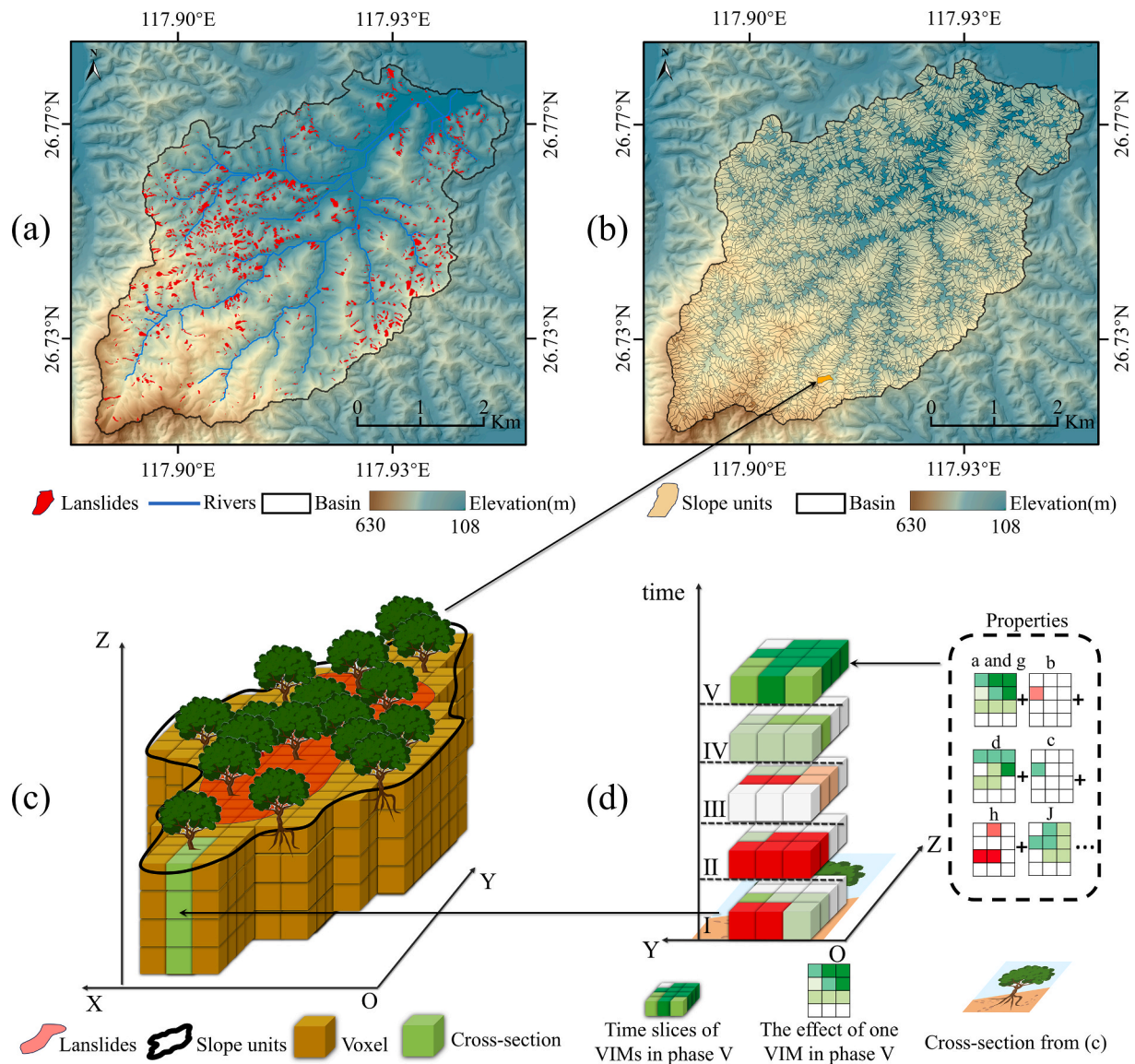


Fig. 7. A case for incorporating vegetation dynamics parameters using space-time cubes. (a) The Caiyuan Basin. (b) Delineation of slope units, where the slope unit used as an example is highlighted. (c) 3D representation of an example slope unit, with the cross-section used as an example highlighted. (d) Time slices of VIMs for the example cross-section under rainfall time series (rainfall time series is divided into phases I, II, III, IV, V, in section 3).

permeability in macropores soils (Beven and Germann, 2013).

In order to apply distributed hydrological models (e.g., TOPMODEL, GEOTOP, SWAT, HYDRUS2D/3D) at the catchment scale, it is necessary to resolve two key issues: firstly, the spatial location distribution of preferential flows and the associated parameters for evaluating their magnitude; and secondly, the value of vertical and lateral hydraulic conductivity at the preferential flow. For hydrological models that calculate overall energy and material fluxes at the catchment scale, there is arguably less need to focus on water transport at the fine scale. Therefore, these suggestions have been validated to some extent by events, but due to the ease of modelling using the Richards equations, this approach continues to be the dominant one. It is imperative that greater attention be paid to the quantification of moisture transport and preferential flow in hydrological models at the FINE scale. For those engaged in landslide mitigation, this will be a prerequisite for the revelation of the spatial and temporal distribution of landslides at the catchment scale.

5.2. Practical considerations for parameters of VIMs in hydrological processes

In addition to the theoretical issues inherent in hydrological models, the incorporating of a number of VIMs parameters into hydrological models remains challenging (Vereecken et al., 2022). Firstly, direct measurements of soil hydraulic properties and vegetation parameters, especially root-related parameters, are often difficult and time consuming. Parameters such as root length density (RLD), root depth (RD), and root area index (RAI) are typically obtained through field surveys and laboratory experiments (Liu et al., 2022). However, such invasive monitoring methods can disrupt the original soil structure, making it difficult to obtain relevant hydraulic parameters. A considerable number of studies will employ probabilistic statistical models with the objective of interpolating vegetation and hydrological parameters obtained at the fine scale to the catchment scale. In contrast, methods like ground penetrating radar (GPR) can provide root

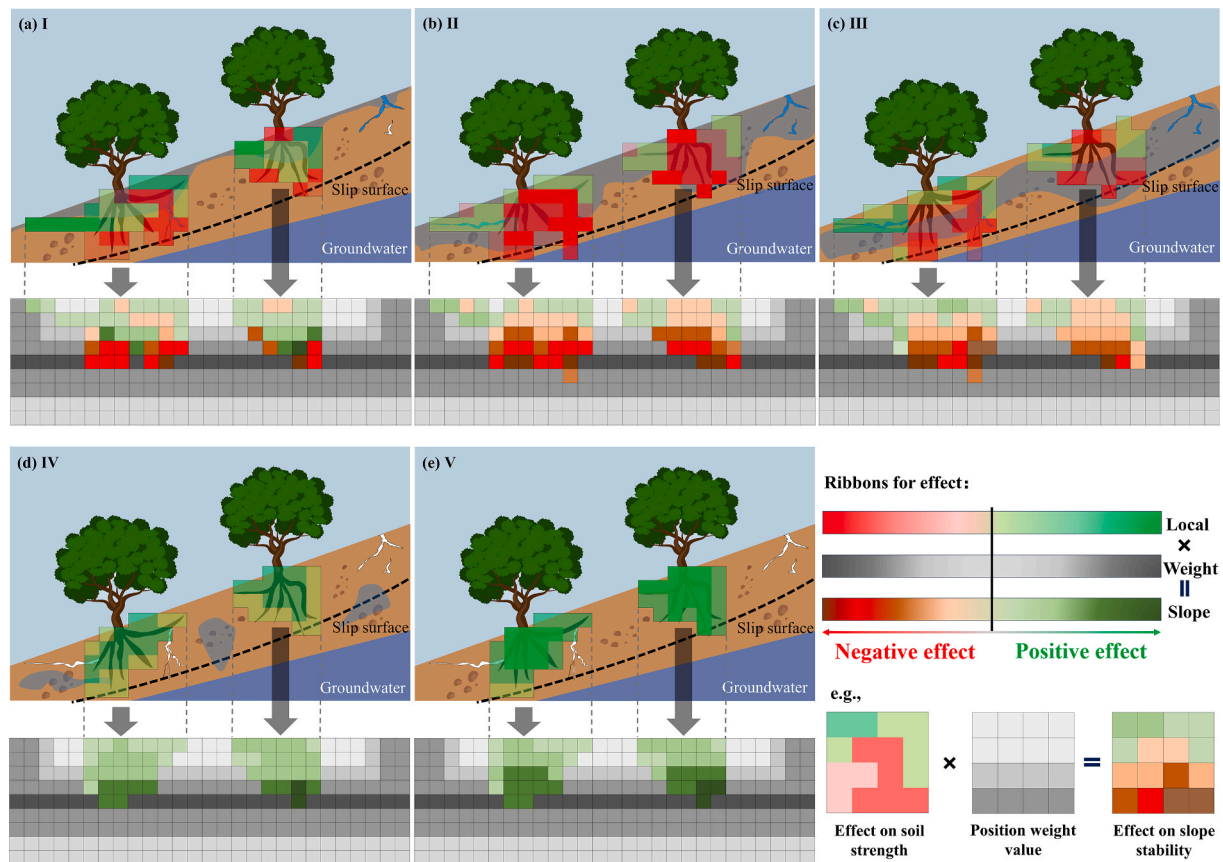


Fig. 8. Scale and magnitude of vegetation hydrological effects on slope stability, they are qualitative expressions for ease of understanding. Colored cubes on the slope represent a specific time slice of the positive and negative effects on soil strength. The composition of these cubes is shown in Fig. 8(b). The grids below represent the importance of slope positions for slope stability. The areas near the potential slip surface, along with the toe and crest of the slope, are considered weak zones. Variations in soil strength in these areas can significantly affect the overall stability of the slope (Gatto et al., 2023; Jiang et al., 2023; Tufano et al., 2021). Therefore, higher weights are assigned to the vicinity of these zones. The colored cubes in the grid, overlaid with colored cubes representing the effect of VIMs on soil strength and gray grids representing positional weights, reflect the true effect of vegetation on slope stability. Due to the presence of the positional weight value, high weight positions result in a darker color when overlaid, while less weight positions, even though they have a significant effect on soil strength, do not have as much effect on slope stability.

parameters at a certain scale while preserving the original soil structure and hydraulic properties, offering promising applications (Raz-Yaseef et al., 2013; Zhu et al., 2014). The main methods to evaluate the effect of RIMs on hydraulic parameters such as soil water retention curve (SWRC), saturated hydraulic conductivity (K_s), and saturated water content (θ_s) include indoor soil column experiments (Scanlan, 2009; Scholl et al., 2014), laboratory tests on field samples (Holtham et al., 2007; Whalley et al., 2005), and in-situ field permeability tests (Fischer et al., 2015; Kodesová et al., 2011). After obtaining the parameters, they are compared with soil hydraulic parameters obtained using PTFs. The reason for quantifying the effect of RIMs on soil hydraulic properties is that conventional PTFs often neglect the influence of roots and microorganisms (Fatichi et al., 2020). However, parameters obtained from indoor soil column experiments or laboratory tests on field samples may not represent soil hydraulic properties at a certain scale, as these methods struggle to account for spatial heterogeneity (Lu et al., 2020; Song et al., 2017). Vegetation stem and leaf parameters, such as transpiration efficiency (TE), leaf area index (LAI), specific leaf area (SLA), and related aboveground hydraulic parameters (e.g., interception loss, stemflow, throughfall) can be obtained through field experiments or indoor cultivation experiments with relevant instrumentation (Gonzalez-Ollauri and Mickovski, 2017a). With the advancement of remote sensing technology, airborne and satellite-based laser scanning can effectively record stem parameters like diameter at breast height (DBH) and plant height, as well as canopy structure parameters,

allowing the estimation of LAI, TE, and other hydraulic parameters that affect vegetation function (Liu et al., 2022).

Furthermore, due to the significant spatiotemporal dependence of vegetation characteristics, distribution, and their effects on hydrological processes during rainfall events (Ghestem et al., 2011), few studies have incorporated VIMs-derived hydraulic parameters into hydrological models. Many models rely on assumptions and simplifications including the nonlinear relationship between root length and water absorption and the neglect of axial flow resistance in roots (Landsberg and Fowkes, 1978; Vereecken et al., 2022). Scanlan (2009) proposed a conceptual model in which roots reduce the effective pore diameter, suggesting that the presence of roots increases capillary suction. However, this model fails to accurately predict preferential flow generated by root channels or changes in soil hydraulic properties caused by cracking. Ng et al. (2016b) developed a soil water retention model considering root penetration, which was further modified and improved (Ni et al., 2019). These studies focus on the spatial positioning of roots in soil pores, while RIMs can increase K_s by inducing additional preferential flow. Some researchers have developed pore network models that consider roots, using X-ray tomography to generate moisture distribution maps and calculate K_s (Holtham et al., 2007; Matthews et al., 2010). In addition, we are able to obtain 2D/3D images of the root system using X-ray and CT scanning technology, which has the advantage of reproducing the real distribution of roots in the soil without destroying the soil structure. Based on the 3D root distribution obtained through CT scanning



Fig. 9. Typical shallow landslide in Yangxi Town, Sanyuan District, Sanming City, Fujian Province, southeast coast of China (Liu, 2022). Unstable slopes and landslides are mainly found in the region where *Phyllostachys edulis* grows. The root system of *Phyllostachys edulis* is characterized by small and densely packed roots, which have limited development of preferential flow and relatively little influence on soil hydrological processes. However, the stems of *Phyllostachys edulis* are slender and closely spaced. Coupled with the presence of special red soils in Fujian Province, which have poor water stability, the additional stress from rainfall, the presence of *Phyllostachys edulis*, and the soil properties are the main factors contributing to slope instability. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

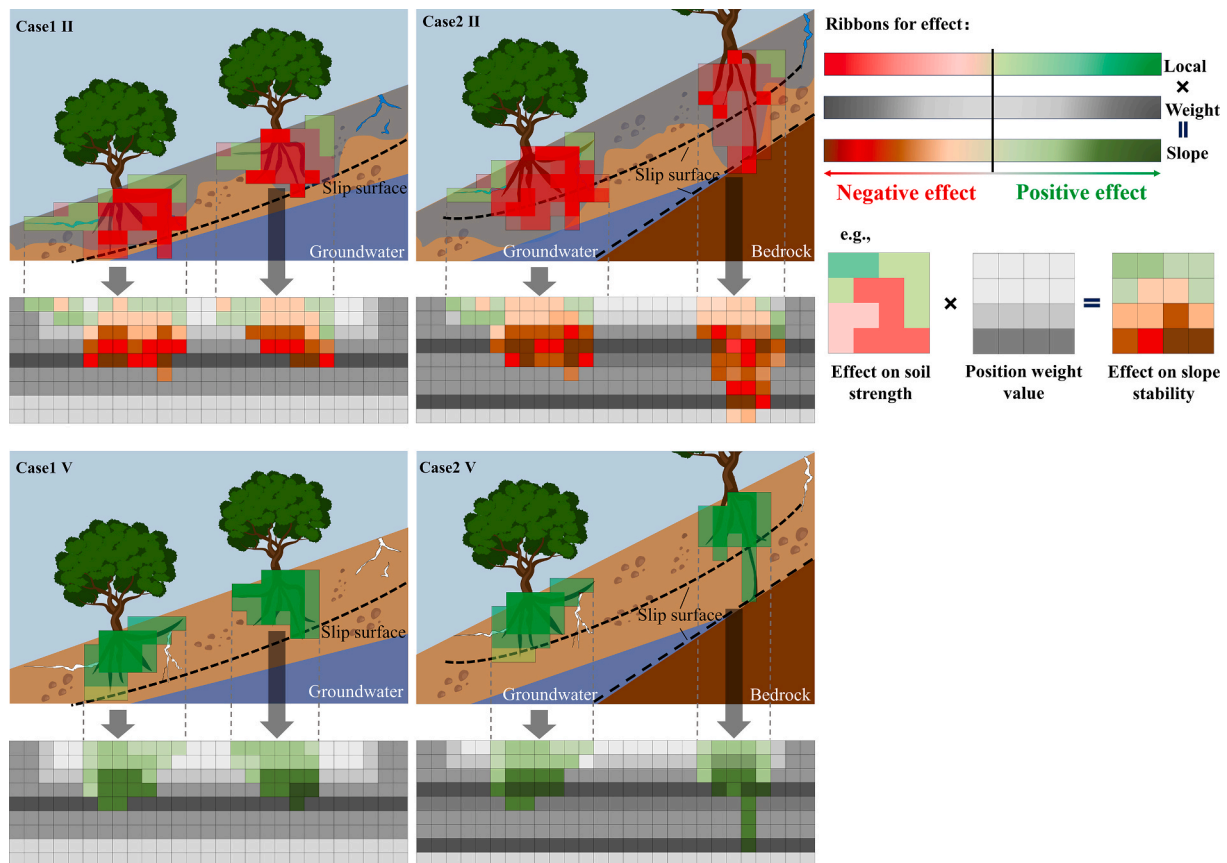


Fig. 10. Comparison of hydrological effects of vegetation during periods of high pore water pressure and high suction on slopes with different characteristics, including the water table, slope, and soil thickness. These are qualitative expressions for ease of understanding. Colored cubes on the slope represent a specific time slice of the positive and negative effects on soil strength. The colored cubes in the grid, overlaid with colored cubes representing the effect of VIMs on soil strength and gray grids representing positional weights. The presence of the positional weight value results in a darker color when high weight positions are overlaid, while lower weight positions, despite exerting a significant effect on soil strength, do not have as much effect on slope stability.

technology, we are able to obtain the RAR, RD and other parameters of the root system through deep learning or other machine learning. Improving the efficiency of parameter acquisition will help to interpolate the parameters from fine scale VIMs to catchment scale.

In future research, we need to emphasize and improve the

understanding of VIMs in hydrological processes and quantify optimal parameter values, and further consider probabilistic statistical models in catchment scale hydrological calculations. Probabilistic statistical models that allow the limited vegetation and soil data to be fully exploited. These form the basis for considering vegetation dynamics

parameters in hydrological models.

5.3. Dynamic slope stability models considering the hydrological effects of VIMs

Slope stability models (SSMs) that account for hydrological and mechanical reinforcement of root systems have been reviewed (Murgia et al., 2022). Both the hydrological and mechanical effects of vegetation on slope stability are highly dynamic in both position and magnitude. Although numerous slope stability models have been developed, they often fail to accurately predict the timing and location of landslides at the catchment scale (Stokes et al., 2014). This is due to oversimplification of highly variable parameters. In most slope stability models, root reinforcement is considered as a uniformly distributed constant cohesion, e.g. GEOTOP-FS (Simoni et al., 2008), TRIGRS (Wu and Sidle, 1995), dSLAM (Baum et al., 2008). To approximate realistic vegetation reinforcement, some models considered the spatial heterogeneity. Mao et al. (2014) distinguishes between vegetated and non-vegetated areas. Arnone associates additional cohesion in vegetated areas with changes in parameters such as root length density (RLD), root radius (RR), and root area index (RAI), further approximating the spatial heterogeneity of vegetated areas (Arnone et al., 2016). Some models even consider the differentiation of root reinforcement mechanisms at the basal and lateral sides of slopes (Cislaghi et al., 2017; van Zadelhoff et al., 2022).

Rainfall interception hydrological models have been reviewed (Muzlyo et al., 2009). Rainfall interception hydrological models that incorporate processes such as interception loss, throughfall, and stemflow (e.g., Gash et al., 1995; Rutter et al., 1971; van Dijk and Bruijnzeel, 2001), have the potential to be combined with hydrological models that consider RIMs to form a comprehensive representation of top-down hydrological processes on slopes. For practical consideration, GEOTOP is a distributed hydrological model considering 3D Richards equations (Rigon et al., 2006), and its advantage of considering lateral infiltration and unsaturated zone pore water pressure variations helps to incorporate the space-time cubes of vegetation hydrological effects. Overall, the extent to which the hydrological model reproduces real hydrological processes, the level of detail in the acquisition of the VIMs parameters, and the optimization of the slope stability analysis are necessary to capture the effects of VIMs in hydrological processes on slope stability and to improve the predictive capability of the distribution of landslides under the catchment scale. Future research should focus on how hydrological models with vegetation influence factors like VIMs are coupled with mechanical parameters such as soil effective stress and additional cohesion under dynamic conditions, ultimately enabling practical application in slope stability models.

6. Conclusion

The phenomenon of global climate change has been accompanied by an increase in the frequency of extreme rainfall events. It is of great importance to gain an understanding of the mechanisms by which vegetation affects soil hydrological processes, and how these effects positively or negatively impact slope stability. This is crucial for the prediction, prevention, and management of rainfall-induced shallow landslides. The impact of vegetation on hydrological processes is multifaceted, with factors such as the funneling effect caused by branches, leaf transpiration, and root compression dynamically influencing these processes.

This review examines various vegetation-induced modifications (VIMs) that affect both internal and surface hydrological processes on slopes. The objective of this review is to summarize the intrinsic relationships between VIMs and hydrological processes and to identify the parameters influencing these processes. Previous research has oversimplified the mechanical mechanisms at the slope scale, failing to consider the impact of vegetation structure on initial infiltration and

preferential flow. Building on this, we discuss how these vegetation hydrological effects, under rainfall conditions, impact local soil strength and overall slope stability.

It is our contention that the influence of VIMs on hydrological processes and the resulting slope stability issues exhibit high spatial and temporal heterogeneity. Dynamic parameter modeling represents an indispensable and non-negligible aspect of future research endeavors. We propose the use of the space-time cube model to analyze the spatiotemporal dynamics and the effects of vegetation on slope stability. This provides a novel theoretical framework for comprehending the intricate interrelationships between vegetation, soil hydrology, and slope stability, enhancing the prediction and management of landslides, and establishing a theoretical foundation for the application of vegetation in slope stabilization.

CRedit authorship contribution statement

Zhichao Deng: Writing – original draft, Methodology, Investigation, Conceptualization. **Hengxing Lan:** Writing – review & editing, Supervision. **Langping Li:** Writing – review & editing, Supervision. **Weifeng Sun:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study is supported by the National Natural Science Foundation of China (Grant No. 42041006 and 41927806), the Fundamental Research Funds for the Central Universities, CHD (Grant No. 300102262901). Authors also want to thank the Editors and the anonymous reviewers for valuable comments and suggestions.

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